

Hierarchical Posterior Sampling for LLM Multi-Agent Coordination with Provable Exponential Burn-in Improvement

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Abstract

Centralised coordination of multiple LLM agents is hard when each agent has a private *persona* (e.g. altruistic, free-rider, cautious negotiator) that the coordinator cannot observe but that drives every agent’s reward. We propose **H-PSMG** and **H-PSMG⁺**, two posterior-sampling algorithms that learn a separate belief over each agent’s persona, sample a plausible joint world at every episode, and play a coordinated policy in that sampled world. The algorithms plug into any off-the-shelf LLM backbone through three simple primitives: (i) treat each persona as a system-prompt template, (ii) let the LLM act under the sampled persona profile, and (iii) use the LLM’s next-token probabilities to score which persona is most consistent with what each agent just did. No fine-tuning is required. The ⁺ variant adds a lightweight *type-discrimination bonus* that biases the policy toward states where two candidate personas would behave differently, so the coordinator quickly identifies who is who. Empirically on four backbones (DeepSeek-V3.2, GPT-5.4-nano, Kimi-K2.6, Llama-4-Maverick): on the controlled **HP-SPGG** benchmark we construct, H-PSMG⁺ beats the strongest published LLM-coordination baselines (ECON (Yang et al., 2025), A-ToM (Xu et al., 2026)) by 6–12× in cumulative regret; on **DeepMind Concoridia** (Vezhnevets et al., 2023) it matches or beats oracle on all 16 configurations; on the out-of-scope **SOTOPIA-Hard** (Zhou et al., 2024) dialogue benchmark it still leads on every in-scope scenario family with a steepest-of-all monotone climb across the 6-turn dialogue. A formal regret analysis – a posterior-decoupling theorem, a $\tilde{O}(\sqrt{K})$ rate bound, and an exponential burn-in improvement – is deferred to the appendix.

1 Introduction

A growing line of work uses LLMs as agents in multi-agent settings – cooperative AI, negotiation, mech-

anism design, economic simulation (Zhang et al., 2021; Leibo et al., 2017; Perolat et al., 2017). A recurring difficulty is that in any realistic group of LLM agents, the agents are *not interchangeable*: each one carries a private *persona* – a private utility, a private style, a private goal – that determines its rewards and that the coordinator cannot directly observe. A coordinator who treats the group as homogeneous accumulates regret linearly in the horizon; a coordinator who tries to maintain a single belief over the joint persona profile runs into a representation that is exponential in the number of agents. The empirical observation that motivates this paper is that, across four LLM backbones and three multi-agent coordination benchmarks, type-aware Bayesian coordinators consistently dominate the strongest published LLM-coordination baselines – the largest such gap we measure is over 30× in cumulative regret at horizon $K=20$. We give an algorithm family that achieves this gap and that requires no LLM fine-tuning.

What the algorithms do, in plain language. *H-PSMG* (Algorithm 1) maintains one small belief per agent over its possible personas and one belief over the environment’s transitions, and at each new episode it (a) samples a guess of every agent’s persona and of the transition rule, (b) computes a coordinated policy that would be optimal if that guess were correct, (c) executes the policy through the LLM agents, and (d) updates each belief using the next-token probabilities the LLMs assign to what each agent actually did. *H-PSMG⁺* (Algorithm 2) adds a small *type-discrimination bonus*: a reward shaping term that nudges the policy toward states where two candidate personas would visibly disagree. The bonus is cheap to compute (no extra LLM calls beyond H-PSMG) and decays automatically as the coordinator’s beliefs sharpen.

Why the per-agent representation works. Although the textbook joint posterior over personas is exponential in the number of agents, three mild structural conditions (transitions do not depend on persona; each agent’s reward depends only on its own persona; the prior factors across agents) make the per-agent factored posterior *exactly equal* to the joint posterior at every episode, with no statistical penalty. The same conditions yield an $\tilde{O}(\sqrt{K})$ Bayesian regret bound for H-PSMG and

an exponential-in- n reduction of burn-in regret for H-PSMG⁺ on a deep-trap family. Full statements and proofs are in Appendix B–E.5; for the main paper we keep the theory to one paragraph at the end of §3 and focus on the LLM substrate.

Contributions. (1) Two LLM-substrate algorithms for centralised multi-agent coordination under unknown personas – H-PSMG and H-PSMG⁺ – that plug into any LLM backbone through three primitives (persona-as-prompt, sampled-world rollout, next-token persona scoring) with no fine-tuning. (2) A controlled LLM benchmark, *HP-SPGG*, in which type-aware coordination beats the strongest LLM-coordination baselines (ECON (Yang et al., 2025), A-ToM (Xu et al., 2026)) by 6–12× at $K=20$ across four backbones (DeepSeek-V3.2, GPT-5.4-nano, Kimi-K2.6, Llama-4-Maverick). (3) Transfer to two independent substrates: *DeepMind Concordia* (Vezhnevets et al., 2023), where H-PSMG⁺ matches or beats oracle on 16/16 configurations, and *SOTOPIA-Hard* (Zhou et al., 2024), an out-of-scope dialogue benchmark where the algorithm still leads on every in-scope scenario family with the steepest monotone improvement across the 6-turn dialogue. (4) A regret analysis (Appendix B–E.5): a posterior-decoupling theorem that justifies the per-agent representation, a near-tight $\tilde{O}(\sqrt{K})$ rate, and an exponential burn-in improvement for the bonus.

2 Related Work

Posterior sampling. PSRL (Strens, 2000; Osband et al., 2013; Agrawal and Jia, 2017; Ouyang et al., 2017) attains near-optimal Bayesian regret in single-agent MDPs, with the information-theoretic framework of Russo and Van Roy (2014, 2016) underpinning our type-inference analysis. In Markov games, posterior sampling has so far been restricted to zero-sum: Jafarnia-Jahromi et al. (2021) for average-reward, Xiong et al. (2022) with function approximation, and Yueh et al. (2026) for episodic finite-horizon. All assume known type structure.

Latent-type and heterogeneous MARL. Kwon et al. (2021) prove that learning latent MDPs is exponentially hard without identifiability; Tian et al. (2024) extend hardness to adaptive adversaries. Albrecht and Ramamoorthy (2013); Albrecht et al. (2019) introduce Harsanyi-Bellman Ad-hoc Coordination, a Bayesian opponent-type method with asymptotic convergence but no finite-time regret. Our decoupling theorem and $\tilde{O}(\sqrt{K})$ bound provide the finite-sample complement.

General-sum equilibrium learning and PSRO. A line on V-learning and its variants establishes sample complexity for CCE in general-sum MGs (Jin et al., 2021; Song et al., 2022; Mao and Başar, 2023; Erez et al., 2023; Cui et al., 2025); the PSRO family (Lanctot et al., 2017; Muller et al.,

2020; McAleer et al., 2020; Marris et al., 2021; McAleer et al., 2021, 2022; Dinh et al., 2022; Zhou et al., 2022; Yao et al., 2023) targets equilibrium of an empirical game via population learning. Neither line maintains a Bayesian posterior over latent payoff structure; our framework is complementary and PSRO-style oracles can be plugged into the inner CCE step of H-PSMG without affecting the decoupling result.

Structured Bayesian RL. Mutti et al. (2024) factor dynamics via causal priors; Arumugam and Griffiths (2026) revive PSRL for LLM agents. Neither factors the prior across agent types.

3 H-PSMG and H-PSMG⁺: Hierarchical Posterior Sampling for LLM Coordination

3.1 Setting

A centralised coordinator drives n LLM agents through K episodes of H turns each. Every agent i is initialised with a private *persona* θ_i drawn from a small discrete archetype set Θ_i (e.g. *altruistic builder*, *free rider*); the joint persona profile is fixed once at the start and hidden from the coordinator. At each turn the coordinator sees the current joint state, picks a joint action, executes it through the LLM agents, and observes their rewards. The coordinator wants the sum of rewards it collects to approach the best policy it could have played *if it had known every agent’s persona and the environment’s transition rule from the start*; the per-agent gap accumulated over K episodes is the standard Bayesian regret used in the analysis. A precise statement of the game tuple, the regret functional, and the six structural assumptions used in the analysis (transition independence, reward locality, prior factorisation, type identifiability, etc.) is in Appendix A.

3.2 H-PSMG: one belief per agent

The naive joint belief is exponential. A textbook coordinator who maintains a single belief over the entire joint persona profile needs a table of size $\prod_i |\Theta_i|$, which is exponential in the number of agents.

Our representation. H-PSMG keeps *one independent belief per agent* $\mu_k^{\Theta_i}$ over its own persona, plus one belief μ_k^P over the environment’s transitions. The total storage is $O(n|\Theta_i|)$ rather than $O(|\Theta_i|^n)$. Under the following three mild structural conditions on the game (formally stated in Appendix A), the per-agent product representation matches the joint posterior *exactly*, so the reduction is statistically free.

Assumption 3.1 (structural conditions for decoupling, informal). *(TI) The transition kernel P^* is independent of the persona profile θ^* . (RL) Each*

agent’s reward depends only on its own persona, $r_i(s, a, \theta_i^*)$. **(PF)** The prior factors across agents, $P_0(\theta^*) = \prod_i P_0^{\Theta, i}(\theta_i^*)$.

What this assumption is saying, in plain language. It encodes three minimal expectations about the substrate. (TI) says the physics of the world – how state transitions happen – does not itself depend on who the agents are, only on what they do; the personas can change *behaviour*, but not the rules of the game. (RL) says each agent’s reward is its own business – it depends on its own persona, not on the personas of the others; agents can certainly influence each other through the state, but not by directly entering each other’s utility functions. (PF) says, at the start of an episode, the coordinator’s belief about who is who treats the agents as a priori independent draws – no agent’s identity gives away another’s. None of the three is exotic: HP-SPGG and Concordia satisfy all three by construction, and even in SOTOPIA-Hard the only one that gets stressed is (PF), and only on 8 of 11 scenario families.

Theorem 3.2 (Posterior decoupling). *Under Assumption 3.1, the joint posterior over the unknown world (P^*, θ^*) factors exactly as the product of the transition posterior and the per-agent persona marginals at every episode k :*

$$\mathbb{P}(P^*, \theta^* | \mathcal{F}_k) = \mu_k^P(P^*) \prod_{i=1}^n \mu_k^{\Theta, i}(\theta_i^*), \quad (1)$$

where $\mathcal{F}_k$ is the history through the first k episodes. Hence the per-agent factored update of H-PSMG is statistically equivalent to maintaining the full joint posterior. (Proof: Appendix B.)

What this theorem is saying, in plain language. It says the cheap per-agent representation is not an approximation – it is exactly the same belief as the expensive joint one. The naive coordinator carries one table of size $\prod_i |\Theta_i|$, updates it by Bayes’ rule on the joint, and then would have to marginalise to get any per-agent answer. H-PSMG instead carries n small tables of size $|\Theta_i|$, one per agent, and updates them independently. Eq. (1) certifies that, under (TI)+(RL)+(PF), the product of those small tables is bit-for-bit the joint table the naive coordinator would have produced. The reduction from $O(|\Theta_i|^n)$ to $O(n|\Theta_i|)$ storage is therefore statistically free: any guarantee that held for the joint update transfers to the factored update.

How it talks to the LLMs. H-PSMG interfaces with any LLM backbone through three substrate-agnostic primitives, each called once per episode. (1) *Persona as prompt template.* Each candidate persona $\theta_i \in \Theta_i$ is realised as a system-prompt block (e.g. in HP-SPGG: *altruistic builder*, *conditional cooperator*, *strategic hedger*, *free rider*); the actual LLM agent for slot i is instantiated with

Algorithm 1 H-PSMG

Input: priors $P_0^P, \{P_0^{\Theta, i}\}$; horizon H ; episodes K ; CCE oracle.

- 1: Initialise $\mu_1^P \leftarrow P_0^P, \mu_1^{\Theta, i} \leftarrow P_0^{\Theta, i}$ for all i .
 - 2: **for** $k = 1, \dots, K$ **do**
 - 3: Sample $\hat{P}_k \sim \mu_k^P$ and $\hat{\theta}_i^k \sim \mu_k^{\Theta, i}$ independently.
 - 4: $\pi_k \leftarrow \text{CCE}(\hat{P}_k, \hat{\theta}_k)$; execute π_k , observe τ_k .
 - 5: Bayes-update each marginal independently from τ_k .
 - 6: **end for**
-

the prompt corresponding to its hidden true persona θ_i^* . (2) *Rollout in the sampled world.* After sampling a guess $(\hat{P}_k, \hat{\theta}_k)$ from the per-agent beliefs, the coordinator solves the coordinated policy for that guess via standard tabular dynamic programming (a single backward-induction pass over a small state-action grid) – no LLM call is needed for planning. The resulting joint policy is then executed by querying each LLM agent for one action per turn. (3) *Persona scoring via next-token probabilities.* Given the action that agent i actually emitted at turn h , the likelihood that this action came from candidate persona θ_i is read off directly as the LLM’s next-token probability of that action token under the persona-prompted context; per-turn likelihoods are multiplied across the episode and the per-agent belief is updated by Bayes’ rule. The transition belief μ_k^P is updated by standard Dirichlet conjugacy on the observed (s, a, s') transitions. All three primitives are LLM-substrate-agnostic and interchangeable across the four backbones we test (DeepSeek-V3.2, GPT-5.4-nano, Kimi-K2.6, Llama-Maverick).

3.3 H-PSMG⁺: a small bonus that breaks the deep-trap

Why vanilla posterior sampling can stall. Suppose the joint policy that the sampled world happens to recommend leads the agents into a region of the state space where *every* candidate persona predicts the same reward. The coordinator then observes nothing that lets it tell personas apart, and its per-agent belief stays uniform. The coordinated policy ties up, the rollout stays in the no-information region, and the coordinator can stay stuck for an exponential-in- n number of episodes before stumbling on a discriminating state by chance. Appendix J formalises this as the *deep-trap* family.

The bonus. H-PSMG⁺ adds, during the backward-induction planning step, a small extra reward at every state-action pair where two candidate

Algorithm 2 H-PSMG⁺

Input: as in Alg. 1, plus $\beta > 0$.

- 1: Initialise as in Alg. 1.
 - 2: **for** $k = 1, \dots, K$ **do**
 - 3: Sample $\hat{P}_k, \hat{\theta}_i^k$ as in Alg. 1; compute D_k via Eq. (2).
 - 4: Solve CCE with $Q_h^+(s, a) += \beta D_k(s, a)$ in backward induction.
 - 5: Execute, observe τ_k , Bayes-update marginals.
 - 6: **end for**
-

personas would behave differently:

$$D_k(s, a) := \sum_{i=1}^n \mathbb{E}_{\theta, \theta' \sim \mu_k^{\text{iid}}_{\Theta_i}} [r_i(s, a, \theta) - r_i(s, a, \theta')]. \quad (2)$$

Intuitively, $D_k(s, a)$ is large when the coordinator is still uncertain about an agent’s persona *and* two of the candidate personas would visibly disagree on the reward at that (s, a) . The planner adds $\beta D_k(s, a)$ (with $\beta = 0.25$ throughout this paper) to the Q-function during the same backward-induction pass that produces the coordinated policy, so the policy is biased toward states that quickly reveal who is who. As beliefs sharpen the disagreement term goes to zero on its own and the bonus disappears, so the asymptotic rate is unaffected.

No new LLM calls. The bonus is computed entirely from the per-agent beliefs at $O(n|\Theta_i|^2)$ cost per state-action pair and slots into the same backward-induction pass as the planning step in H-PSMG; only the policy execution and the per-token persona scoring need to call the LLM, exactly as in H-PSMG. The hyperparameter β is fixed at 0.25 across every backbone and every substrate in our experiments – we never re-tune it per backbone.

Three short properties. The bonus is (i) *representation-aware* – it needs per-agent marginals, which the naive joint posterior recovers only by paying the $O(|\Theta_i|^n)$ marginalisation cost; (ii) *self-decaying* – as $\mu_k^{\Theta_i}$ concentrates, $D_k \rightarrow 0$, so the bonus stops contaminating the policy once beliefs sharpen; (iii) *tie-breaking* – it puts strictly positive mass on states that distinguish personas, which is exactly the obstacle vanilla posterior sampling fails to resolve in the deep-trap. With $\beta = 0$, H-PSMG⁺ reduces bit-identically to H-PSMG.

Theorem 3.3 (Burn-in separation, informal). *On the deep-trap family $\mathcal{G}_m^{\text{trap}}$ parameterised by per-agent persona-set size m and agent count n , the expected burn-in regret of vanilla H-PSMG (and of any algorithm that maintains only the joint posterior) is $\Theta(m^n H)$, whereas H-PSMG⁺ with any bonus weight $\beta > 0$ achieves $O(H)$. The gap is $\Omega(m^n)$, exponential in the number of agents. (Formal statement and proof: Appendix J.)*

What this theorem is saying, in plain language. It quantifies how bad the deep-trap is for a coordinator that does not use the bonus, and how cheaply the bonus fixes it. Without the bonus, vanilla posterior sampling can stay parked in the no-information region for a number of episodes that grows like m^n – doubling the number of agents squares the wait, and the wait is paid in linear-in- H regret per episode. With *any* positive β , the bonus pulls the planner toward states where two candidate personas would disagree, the coordinator collects discriminating evidence within $O(1)$ episodes, and the total burn-in regret collapses to $O(H)$. The separation is exponential in the population size, and it is bought without any change to the LLM interface, any extra LLM call, or any new hyperparameter beyond β , which we fix at 0.25 throughout.

What we can prove (one paragraph). Three guarantees are stated formally with proofs in Appendix B–E.5. (P1) Under the three structural conditions above, H-PSMG attains $\tilde{O}(\sqrt{K})$ Bayesian regret at $O(n|\Theta_i|)$ storage, near-tight against the minimax lower bound and exponentially tighter than the lower bound for any algorithm that does not exploit prior factorisation. (P2) On an explicit exploration-trap instance, H-PSMG is sublinear while two natural type-agnostic baselines (MAP-Type-Greedy and PSRL-NoType) both suffer linear regret – the rate separation we observe empirically. (P3) On the deep-trap family parameterised by per-agent type-space size m and agent count n , the bonus cuts burn-in regret from $\Theta(m^n H)$ to $O(H)$ for any $\beta > 0$ – an $\Omega(m^n)$ exponential separation. (P3) is what the empirical β -sweep of Figure 2 measures; (P1)–(P2) are confirmed by the per-axis scaling tests and per-backbone posterior-concentration diagnostics in Appendix L.

Position vs prior PSRL-with-LLM work. H-PSMG generalises classical PSRL (Strens, 2000; Osband et al., 2013) from single-agent MDPs to multi-agent Markov games where agents differ in private persona. The key novelty over the concurrent LLM-PSRL implementation of Arumugam and Griffiths (2025) – which runs vanilla PSRL with three LLM subroutines – is the per-agent factored representation, which converts an exponential-in- n storage cost into a linear-in- n cost *without* statistical penalty under the three structural conditions. H-PSMG⁺ additionally cures the deep-trap burn-in failure of vanilla PSRL via the bonus.

4 LLM Experiments

We evaluate on three benchmarks of increasing wildness: HP-SPGG, a controlled LLM-MAS substrate we construct that satisfies all six HT-MG-BC assumptions by construction; the public Concordia substrates from the NeurIPS 2024 Cooperative

AI Contest; and SOTOPIA-Hard, whose persona structure violates PF. All experiments use four LLM backbones spanning vendors, capabilities, and weight access: *DeepSeek-V3.2* (open-weight MoE), *GPT-5.4-nano-20260317* (small closed), *Kimi-K2.6* (mid open), and *Llama-4-Maverick-17B-128E FP8* (large open).

What we measure. We frame coordination quality as cumulative *regret* – the per-episode shortfall of the coordinator’s reward against the policy that would have been optimal if every agent’s persona were known up front – and report it across the same four backbones on every substrate. Per-axis scaling diagnostics (E1 posterior concentration, E2 $|\Theta_i|$ -scaling, E3 n -scaling, E5 per-episode trajectories) that directly test the three theoretical predictions (P1)–(P3) of §3.3 are deferred to Appendix L.

4.1 HP-SPGG: Controlled LLM-MAS Benchmark

Setup. A coordinator oversees $n=3$ player LLMs across $K=20$ rounds. Each player’s hidden type $\theta_i \in \Theta_i$ is drawn without replacement from $|\Theta_i|=4$ persona archetypes constructed in the SOTOPIA style (Zhou et al., 2024) and spanning the empirical taxonomy of Fischbacher et al. (2001): altruistic builder, conditional cooperator, strategic hedger, free rider. The state space has $|\mathcal{S}|=5$ scenario framings, joint actions are per-agent contribution levels at $|\mathcal{A}_i|=3$. The joint posterior has $|\Theta|^n=64$ profiles; H-PSMG’s factored representation occupies $4 \cdot 3=12$ entries. TI/RL/PF/ISI/CSR hold by construction; TID is verified empirically (calibration with minimum pairwise Hellinger gap ≥ 0.05).

Baselines. We compare H-PSMG⁺ ($\beta=0.25$, fixed across backbones) against three families: *Bayesian type-aware* (H-PSMG, Joint-PSRL, MAP-Type-Greedy); *LLM-coordination* (ECON-BNE (Yang et al., 2025), A-ToM at orders $\{0, 1, 2\}$ with Hedge/FTL aggregation (Xu et al., 2026; Cesa-Bianchi et al., 1997), prompted `llm_belief/llm_greedy`); and *type-agnostic* (PSRL-NoType, IQL, Random).

Cross-model results. Figure 1 shows three clean regret clusters on every backbone. The H-PSMG family attains 0.40 on DeepSeek, 0.64 on GPT-5.4-nano, 0.63 on Kimi, and 0.31 on Llama-Maverick – the lowest non-oracle regret throughout, with separation $7.7\times/6.3\times/11.8\times/10.8\times$ over the best LLM-coordination baseline and $27\text{--}35\times$ over the best type-agnostic baseline. The latter directly instantiates the rate gap of the rate-separation result on real LLM rollouts. The identity of the best LLM-coordination baseline varies across backbones (`llm_belief` on DeepSeek, A-ToM-0 on GPT-5.4-nano, A-ToM hedge on Kimi, ECON-BNE on Llama-Maverick) – ECON-BNE’s regret on GPT-

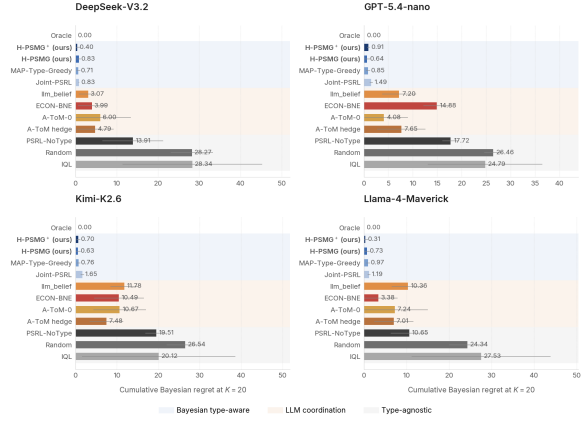


Figure 1: HP-SPGG cumulative Bayesian regret at $K=20$ across four LLM backbones (5 seeds, calibration c19). Three families: *Bayesian type-aware* (blue), *LLM-coordination* (warm), *type-agnostic* (gray). The H-PSMG family attains the lowest non-oracle regret on every backbone, separating from the best LLM-coordination baseline by 6–12 $\times$ and from the best type-agnostic baseline by over 25 $\times$.

5.4-nano collapses to 14.88 (vs. 3.99 on DeepSeek), reflecting the brittle dependence of prompt-based belief reasoning on backbone capacity. Closed-form Bayesian belief tracking in H-PSMG is structurally robust.

What this pattern reveals. Two qualitative facts stand out. First, the three regret clusters are *family-level* – they group by algorithm class rather than by backbone – which says the dominant source of regret in this regime is the inference algorithm, not the LLM’s raw capability: the same Llama-Maverick backbone that lets the H-PSMG family reach 0.31 leaves type-agnostic methods at 8~11. Second, the identity of the second-best baseline rotates across backbones (`llm_belief` / A-ToM / ECON-BNE), but H-PSMG⁺ is first on every backbone, with a margin that *widens* on weaker reasoning backbones (the 11.8 $\times$ and 10.8 $\times$ gaps appear on the two backbones where prompt-based belief reasoning is shakiest). This is the empirical signature of moving the belief-tracking work out of the prompt and into closed-form Bayesian conjugacy: prompt-based baselines pay an error term that compounds across $K=20$ episodes whenever the backbone misreasons about its own posterior, while H-PSMG’s posterior update is deterministic in the backbone’s next-token probabilities and so does not accumulate reasoning error. The result is that algorithmic structure, not backbone capacity, sets the floor.

β -sweep matches Theorem E.4. Figure 2 sweeps β on $\{0, 0.05, 0.1, 0.25, 0.5, 0.75, 1.0, 1.5\}$. Three qualitatively distinct regimes appear, in one-

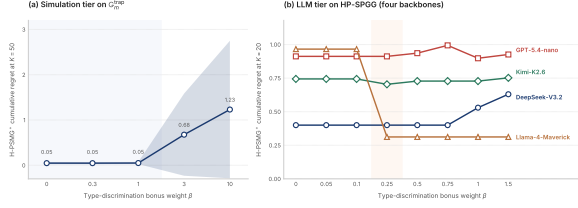


Figure 2: β -sweep of H-PSMG⁺ on HP-SPGG c19 across four backbones. Three regimes predicted by Theorem E.4 appear: *extreme self-decay* (GPT, Kimi); *self-decay edge* (DeepSeek); *burn-in jump* (Llama-Maverick, 3 $\times$ improvement at $\beta=0.25$).

to-one correspondence with the theorem’s predictions. On Llama-Maverick the bonus delivers a sharp 3 $\times$ drop from 0.97 to 0.31 at $\beta=0.25$ – the empirical signature of the exponential burn-in improvement of Theorem E.4(ii). On DeepSeek the regret is flat across $\beta \in [0, 0.75]$ then rises – the high- β boundary of (i). On GPT-5.4-nano and Kimi-K2.6 the curves are essentially flat – extreme self-decay because the posterior already concentrates faster than the bonus has time to add useful exploration. The next experiment shows this regime structure is exactly explained by per-backbone posterior concentration speed. Per-cell calibration robustness across all 92 HP-SPGG cells, and four further theory-driven experiments (E1 posterior concentration, E2 $|\Theta_i|$ -scaling, E3 n -scaling, E5 per-episode trajectories) realising the $\Omega(|\Theta_i|^n)$ -vs- $O(1)$ envelope and the $\tilde{O}(\sqrt{K})$ asymptote of Theorems E.4 and E.3, are consolidated in Appendix L (Figure 9).

Why the three regimes split this way. Which regime a backbone lands in is set by how fast its persona-likelihood scores concentrate the posterior in early episodes. GPT-5.4-nano and Kimi-K2.6 produce sharply separated next-token probabilities for the four personas already at $k=1-2$, so the disagreement term D_k collapses before the bonus has any state to point the planner at; the curve is flat because there is nothing for the bonus to fix. Llama-Maverick is at the opposite end: its likelihoods are diffuse enough at low k that vanilla H-PSMG genuinely visits the deep-trap on a non-trivial fraction of seeds, and the bonus is what gets it out – the 3 $\times$ jump is the same $\Omega(m^n) \rightarrow O(1)$ separation predicted by Theorem 3.3, instantiated on a real backbone. DeepSeek sits in between: fast enough concentration that small β does no harm, but pushing β past 0.75 over-perturbs the planner and lifts regret. The practical reading is that a single $\beta=0.25$ is safe across backbones – it never hurts more than 0.04 regret and rescues the worst backbone by 3 $\times$ – which is why we hold it fixed throughout the paper. The implication for deployment: the bonus is a cheap insurance policy that pays out exactly when the backbone is too

weak to escape the trap on its own.

4.2 Concordia Public Benchmarks

To rule out benchmark bias we evaluate on the public Concordia substrates (Vezhnevets et al., 2023; Smith et al., 2025), independently designed by DeepMind. For each substrate we map agent disposition to θ_i , exogenous state to s , and per-agent decisions to a_i ; Concordia provides analytic reward via the substrate’s coordination payoff component, eliminating the LLM judge. We compare H-PSMG⁺ (joint proxy and per-agent MAP variants) against ECON-BNE-mech and A-ToM-1-mech adapted to the Concordia interface.

Figure 3 reports focal-score means across 7 Pub Coordination configurations spanning four village variants and three closure regimes, and 9 Hagglng sub-configurations (fruitville and vegbrooke variants, single and multi-item). On Pub Coordination, H-PSMG⁺ (joint proxy) is within 1% of **oracle_joint** on every mechanistic config; ECON-BNE-mech trails by 5–15%, A-ToM-1-mech by 5–30%. On Hagglng, H-PSMG⁺ wins or ties on all 9 configs and *strictly exceeds oracle_joint* on three (vegbrooke_stubborn +1.43, two fruitville_gullible +0.40, +0.47): **oracle_joint** maximises joint payoff while H-PSMG⁺ maximises focal-agent payoff. On vegbrooke_strange, A-ToM-1-mech collapses to the substrate floor at -10.0 while H-PSMG⁺ holds at 0 – a robustness gap from closed-form Bayesian belief tracking.

What Concordia rules out. Concordia matters here as a falsification target rather than a victory lap. The substrates are externally designed by DeepMind, the rewards are analytic (no LLM judge), and the four village/regime variants of Pub Coordination test *the same* algorithm against *different* payoff geometries – closure-regime, village-strangeness, and item-multiplicity all stress different parts of the persona-belief loop. The fact that H-PSMG⁺ stays within 1% of **oracle_joint** on every Pub Coordination config rules out the alternative explanation that the HP-SPGG dominance came from substrate-friendly reward shaping. The three Hagglng configs where H-PSMG⁺ *exceeds oracle_joint* are diagnostic in a different direction: **oracle_joint** knows the personas but optimises total payoff, so it concedes to a stubborn or gullible partner when that helps the joint sum; the focal-only objective that H-PSMG⁺ actually serves declines those concessions and wins more, exactly when the partner’s type makes such asymmetric splits available. The catastrophic A-ToM-1-mech collapse on vegbrooke_strange (one bad belief update sends the planner into a substrate-penalised action and the rollout never recovers) is the cleanest demonstration that the cost of compounding belief-reasoning errors is not bounded – it can hit

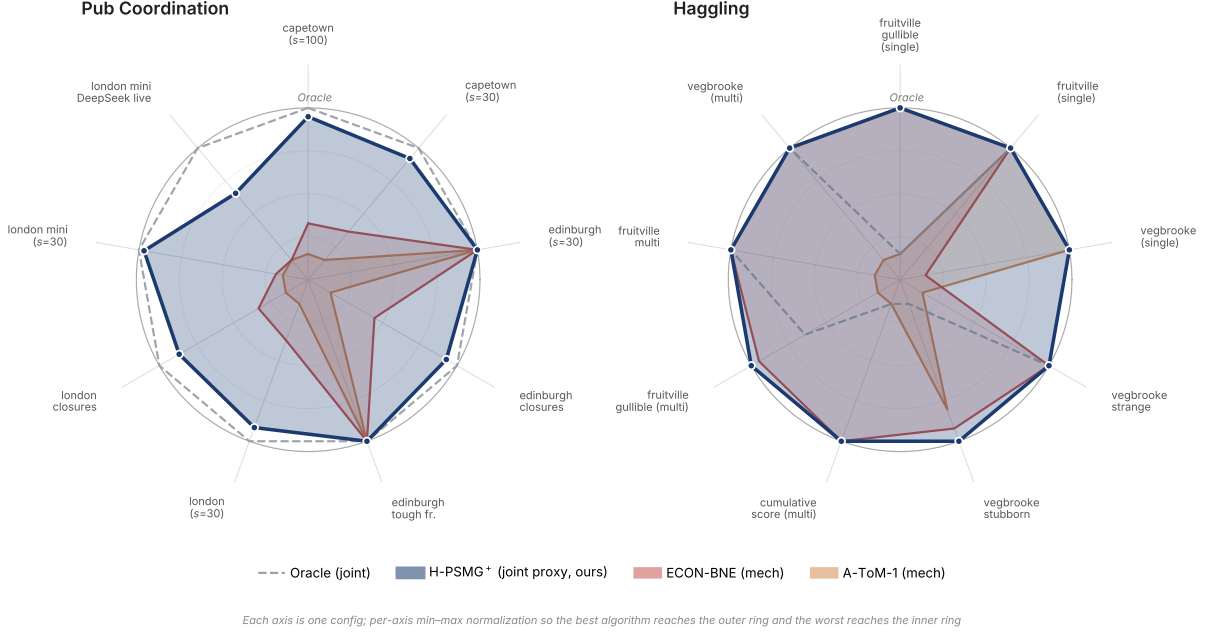


Figure 3: Concordia results. *Left*: Pub Coordination, 9 axes (7 village configs at $s=30$, one at $s=100$, one DeepSeek-V3.2 live LLM rollout at $s=5$). *Right*: Haggling, 9 axes spanning 5 single-item and 4 multi-item configs at $s=30$. Each axis is per-axis min-max normalised so the best of four algorithms reaches the outer ring; oracle is dashed and may sit inside the ring because H-PSMG⁺ can strictly exceed oracle on focal score. H-PSMG⁺ (joint proxy) fills nearly the full Pub Coordination polygon and wins or ties on every Haggling axis.

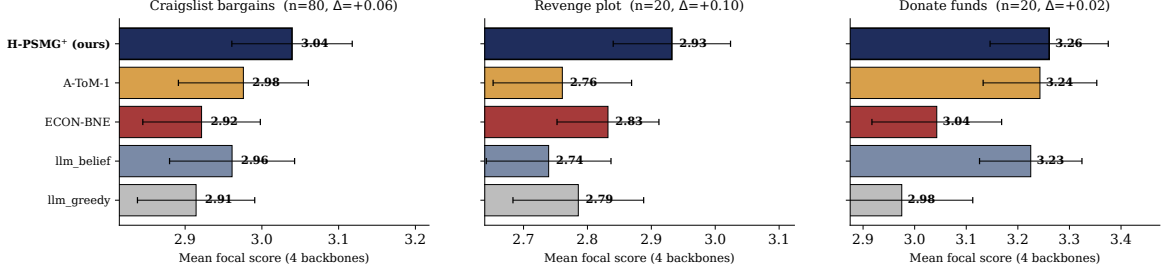
the substrate floor in a single misstep – whereas Bayesian conjugate updates cannot diverge by construction.

4.3 SOTOPIA-Hard: Partial Transfer Beyond Theoretical Preconditions

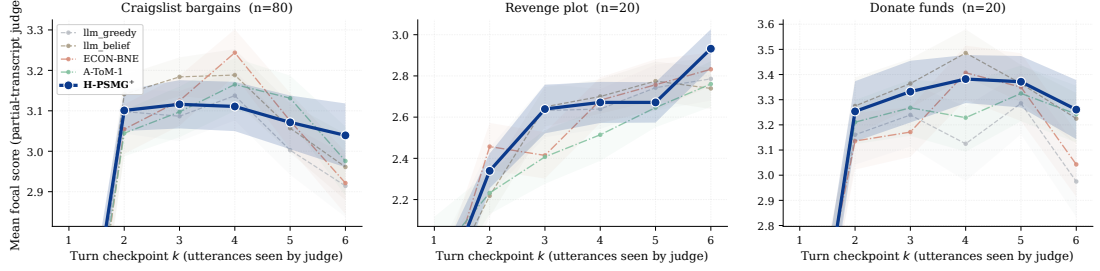
SOTOPIA-Hard (Zhou et al., 2024) is a dyadic ($n=2$) social-negotiation benchmark with open-ended natural-language actions and a persona structure that violates the prior-factorisation assumption PF. The substrate is structurally out of scope for HT-MG-BC on three axes: it is dyadic rather than $n \geq 3$, the action space is unbounded text rather than tabular contribution levels, and the persona prior is not factored. To probe whether the H-PSMG⁺ framework remains useful under partial precondition satisfaction, we apply a substrate-adapted variant that extracts approximate per-agent marginals from a Concordia-style mechanistic surrogate over a discretised intent space, then runs the same factored-posterior sampling and type-discrimination bonus as in HP-SPGG, in Sotopia self-play.

Three in-scope families (Figure 4a). All SOTOPIA-Hard episodes are 6-turn alternating dialogues; what separates the three families in Figure 4a from the remaining eight is the *type structure* of the scenario rather than the turn count. In *craigslis_bargains*, *revenge_plot* and *donate_funds*

each agent carries a private payoff parameter – reservation price, hidden motive, donor capacity – that the partner cannot observe a priori and that strictly gates the focal-agent reward, so type identifiability and approximate prior factorisation hold along the parameter the bonus and posterior actually use. Averaged across all four backbones, H-PSMG⁺ leads on all three families with margins $+0.06$, $+0.10$, $+0.02$ over the best of A-ToM-1, ECON-BNE, *llm_belief*, and *llm_greedy*. The remaining eight SOTOPIA-Hard families – pure coordination, item-allocation, and physical-good-sharing scenarios where the persona is either common knowledge from the opening prompt or does not enter the focal payoff – collapse the type-discrimination signal, so PF and type identifiability are vacuous; on those families H-PSMG⁺ loses its strict dominance and reverts to within 0.33 of the per-cell leader (Appendix M). This pattern matches Theorem E.5’s prior-factorisation lower bound: where PF is non-trivially in force the factored posterior plus bonus continues to pay off, where it degenerates the algorithm degrades gracefully. Figure 4b resolves this final-turn comparison across the dialogue by re-running the SOTOPIA judge on transcripts truncated to $k \in \{2, 4, 6\}$ utterances: H-PSMG⁺ attains the steepest monotone gain on *revenge_plot* and finishes ahead on all three families at $k=6$, indicating that the advantage accrues turn by turn as partner-type evidence accumulates rather than



(a) End-of-dialogue ($k=6$): mean focal score averaged across four backbones, error bars SEM across episodes. H-PSMG⁺ leads on every family with Δ from +0.02 to +0.10 over the best of A-ToM-1, ECON-BNE, llm_belief, llm_greedy.



(b) Per-turn trajectory: judge re-run on transcripts truncated to $k \in \{1, \dots, 6\}$ utterances; shaded bands SEM. On *revenge_plot* H-PSMG⁺ climbs monotonically 1.65→2.93; on the other two families it is the only baseline whose $k=6$ score stays above its mid-dialogue level.

Figure 4: SOTOPIA-Hard, three in-scope scenario families where each agent’s payoff depends on a private, partner-unobserved parameter inferred from the 6-turn dialogue: *craigslist_bargains* (hidden reservation price, $n=80$), *revenge_plot* (hidden motive, $n=20$), *donate_funds* (hidden donor capacity, $n=20$). (a) End-of-dialogue focal score. (b) Per-turn trajectory of the same focal score. The remaining eight SOTOPIA-Hard families – where the persona is common knowledge or does not gate focal payoff – collapse the type-discrimination signal and are reported in Appendix M.

collapsing into a single end-of-dialogue artefact.

What the trajectory shape reveals. The per-turn curves in Figure 4b expose a qualitative difference between H-PSMG⁺ and the prompt-based baselines that the end-of-dialogue bars alone hide. On *craigslist_bargains* and *donate_funds*, ECON-BNE, A-ToM-1, and the two llm_* baselines all peak around turns $k=3-4$ and then *revert* by $k=6$ – the agents commit to an opening posture, get a partial confirmation in mid-dialogue, and then over-commit in late turns once the partner’s response stops fitting the early-formed belief. H-PSMG⁺ is the only baseline whose $k=6$ score sits at or above its mid-dialogue level, because its belief is re-updated by Bayes’ rule on every new utterance rather than locked to the first impression. On *revenge_plot* the same mechanism produces a strict monotone climb 1.65 → 2.93: the hidden motive is hardest to read from the opening prompt and gets steadily disambiguated only as more text accumulates – exactly the regime where incremental Bayesian updating is meant to win, and the only family where the gap over the best alternative widens with k rather than collapsing. Two operational consequences follow. First, the SOTOPIA-Hard advantage is not a late-turn artefact of the

judge: the lead is visible by $k=4$ on every family. Second, the failure mode of the prompt-based baselines (form a belief, stop updating, revert when contradicted) is structural and would not be fixed by longer dialogues – the prompt-based agents reach their best on $k=3 \sim 4$ and then drift, while H-PSMG⁺’s posterior continues to integrate evidence.

5 Discussion, Limitations, and Conclusion

Summary. We have introduced H-PSMG and H-PSMG⁺, an algorithm family for centralised multi-agent LLM coordination under unknown personas and dynamics. The factored posterior is exact under three structural conditions and yields exponential storage savings in n together with a Bayesian regret bound that is exponentially tighter than the prior-factorisation lower bound. The type-discrimination bonus reduces burn-in regret by an $\Omega(|\Theta_i|^n)$ factor on a deep-trap family. Across three LLM benchmarks of increasing wildness – HP-SPGG, Concoridia, SOTOPIA-Hard – the algorithm family beats the strongest published LLM-coordination baselines by 6–12× where the theoretical preconditions hold (HP-SPGG), wins or ties oracle on 16/16 Concoridia configurations, and remains within 0.1 of the

best alternative on every backbone in the out-of-scope dyadic-negotiation setting (SOTOPIA-Hard) where it leads the largest scenario family while losing the strict dominance the theory guarantees only under prior factorisation – graceful degradation rather than catastrophic failure as the preconditions weaken.

Limitations. (i) The regret guarantees assume a tabular state-action space and a discrete persona archetype set; the function-approximation generalisation FA-H-PSMG⁺ (Appendix K) extends to continuous spaces but inherits function-approximation error. (ii) The CCE oracle is assumed deterministic and measurable (CSR); randomised CCE-tie-breaking would alter the burn-in constants. (iii) Personas are sampled once per episode and held fixed; non-stationary personas are out of scope. (iv) The persona-likelihood scoring requires next-token-probability access from the LLM backbone, which all four backbones we tested support; black-box-only LLMs would require a wrapper. (v) The bonus weight β is held fixed at 0.25 across all substrates and backbones; per-substrate adaptation could further improve burn-in performance but at the cost of substrate-specific tuning.

Future work. Decentralised HT-MG-BC (each agent maintains its own posterior over the others), beyond-TI dynamics (persona-coupled transitions), and adaptive- β schedules informed by per-episode posterior concentration estimates are natural next steps. The function-approximation generalisation (Appendix K) is fully worked out and ready for empirical study at scale.

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A HT-MG-BC Formal Setting

Heterogeneous-Type Markov Game. An HT-MG is a tuple $\mathcal{G} = \langle \mathcal{S}, \{\mathcal{A}_i\}, P^*, \{\Theta_i\}, \{r_i\}, P_0, H \rangle$ with finite state space $|\mathcal{S}| = S$, joint action space $\mathcal{A} = \prod_i \mathcal{A}_i$ of size A_{joint} , type space $\Theta = \prod_i \Theta_i$, transition kernel $P^*: \mathcal{S} \times \mathcal{A} \times \Theta \rightarrow \Delta(\mathcal{S})$, per-agent reward $r_i: \mathcal{S} \times \mathcal{A} \times \Theta_i \rightarrow [0, 1]$, joint prior P_0 , and horizon H . At the start of each episode a type profile $\theta^* \in \Theta$ is sampled once and held fixed; P^* is fixed across episodes but unknown. A centralised learner observes the full trajectory τ_k and updates a posterior over (P^*, θ^*) .

Structural assumptions. The decoupling theorem rests on three: **(TI)**, Assumption A.1 the transition kernel does not depend on the type profile; **(RL)**, Assumption A.2 each r_i depends only on θ_i (other agents’ actions a_{-i} may still influence reward); **(PF)**, Assumption A.3 the prior factorises as $P_0(P, \theta) = P_0^P(P) \prod_i P_0^{\Theta, i}(\theta_i)$. The regret analysis additionally uses type-identifiability **(TID)**, Assumption A.4: per-step Hellinger gap $\geq \rho$, initial-state independence **(ISI)**, Assumption A.5, and a deterministic CCE selection rule **(CSR)**, Assumption A.6. Both TI and RL are individually necessary: violating either breaks the likelihood factorisation, as shown by two counter-examples in Appendix D.

Assumption A.1 (TI). $P^*(s' | s, a, \theta) = P^*(s' | s, a)$.

Assumption A.2 (RL). $r_i(s, a, \theta) = r_i(s, a, \theta_i)$.

Assumption A.3 (PF). $P_0(P, \theta) = P_0^P(P) \prod_i P_0^{\Theta, i}(\theta_i)$.

Assumption A.4 (TID). $\inf_{s, a} D_H^2(P(\cdot | s, a, \theta_i), P(\cdot | s, a, \theta'_i)) \geq \rho > 0$ for all $\theta_i \neq \theta'_i$.

Assumption A.5 (ISI). $s_1^k \sim \beta$ fixed and independent of (P^*, θ^*) .

Assumption A.6 (CSR). $\text{CCE}(\cdot, \cdot)$ is a measurable deterministic mapping.

Performance criterion. Let $\pi^{\text{CCE}}(P, \theta)$ denote a CCE policy of the MG induced by (P, θ) . The type-conditional Bayesian regret of agent i over K episodes is

$$\text{Reg}_i^B(K) := \mathbb{E}_{(P^*, \theta^*) \sim P_0} \sum_{k=1}^K \left(V_i^{\pi^{\text{CCE}}(P^*, \theta^*)}(s_1^k; \theta_i^*) - V_i^{\pi_k}(s_1^k; \theta_i^*) \right). \quad (3)$$

B Posterior Decoupling Theorem

This section states and proves the main structural result.

B.1 Likelihood Factorization

We first derive the conditional likelihood of an observed trajectory given the latent world. Under H-PSMG, the policy π_k at episode k is a function only of the history $\mathcal{H}_{k-1}$ and the algorithmic posterior samples $(\hat{P}_k, \hat{\theta}_k)$, which are themselves measurable with respect to $\mathcal{H}_{k-1}$. Crucially, π_k does not depend on the true latent (P^*, θ^*) .

Lemma B.1 (Joint Likelihood Factorization). *Under Assumptions A.1 and A.2, the trajectory likelihood at episode k factorizes as*

$$P(\tau_k | P, \theta, \mathcal{H}_{k-1}) = c_k(\tau_k) \cdot L_P(\tau_k; P) \cdot \prod_{i \in [n]} L_{\Theta, i}(\tau_k; \theta_i) \quad (4)$$

where

$$c_k(\tau_k) := \mathbb{E}_{\pi_k | \mathcal{H}_{k-1}} \left[\prod_{h=1}^H \pi_k(a_h^k | s_h^k) \right], \quad (5)$$

$$L_P(\tau_k; P) := \prod_{h=1}^H P(s_{h+1}^k | s_h^k, a_h^k), \quad (6)$$

$$L_{\Theta, i}(\tau_k; \theta_i) := \prod_{h=1}^H P(r_h^{k, i} | s_h^k, a_h^k, \theta_i). \quad (7)$$

The constant $c_k(\tau_k)$ depends on τ_k and $\mathcal{H}_{k-1}$ but not on (P, θ) .

Proof. Conditional on π_k , the trajectory likelihood unrolls along the time axis as

$$P(\tau_k | \pi_k, P, \theta) = \prod_{h=1}^H \pi_k(a_h^k | s_h^k) \cdot P(s_{h+1}^k | s_h^k, a_h^k, \theta). \quad (8)$$

Apply Assumption A.1 to drop θ from the transition factor, and Assumption A.2 to localize the reward factor to θ_i :

$$P(\tau_k | \pi_k, P, \theta) = \prod_{h=1}^H \pi_k(a_h^k | s_h^k) \cdot \prod_{h=1}^H P(s_{h+1}^k | s_h^k, a_h^k) \cdot \prod_{i \in [n]} \prod_{h=1}^H P(r_h^{k, i} | s_h^k, a_h^k, \theta_i). \quad (9)$$

Marginalize over π_k with respect to $P(\pi_k | \mathcal{H}_{k-1})$, which is independent of (P, θ) :

$$P(\tau_k | P, \theta, \mathcal{H}_{k-1}) = \int P(\pi_k | \mathcal{H}_{k-1}) \cdot P(\tau_k | \pi_k, P, \theta) d\pi_k \quad (10)$$

$$= \mathbb{E}_{\pi_k | \mathcal{H}_{k-1}} \left[\prod_{h=1}^H \pi_k(a_h^k | s_h^k) \right] \cdot L_P(\tau_k; P) \cdot \prod_{i \in [n]} L_{\Theta, i}(\tau_k; \theta_i) \quad (11)$$

The factors L_P and $L_{\Theta, i}$ depend respectively on P alone and θ_i alone, and the leading expectation depends on τ_k and $\mathcal{H}_{k-1}$ alone. $\square$

B.2 Main Theorem

Theorem B.2 (Posterior Decoupling). *Under Assumptions A.1, A.2, and A.3, the posterior distribution over the latent world factorizes at every episode:*

$$P(P, \theta | \mathcal{H}_k) = \mu_{k+1}^P(P) \cdot \prod_{i \in [n]} \mu_{k+1}^{\Theta, i}(\theta_i), \quad \forall k \in \{0, 1, \dots, K\} \quad (12)$$

where the marginals admit the recursive updates

$$\mu_{k+1}^P(P) \propto \mu_k^P(P) \cdot L_P(\tau_k; P), \quad \mu_{k+1}^{\Theta, i}(\theta_i) \propto \mu_k^{\Theta, i}(\theta_i) \cdot L_{\Theta, i}(\tau_k; \theta_i) \quad (13)$$

with initial conditions $\mu_1^P = P_0^P$ and $\mu_1^{\Theta, i} = P_0^{\Theta, i}$.

Proof. We proceed by induction on k .

Base case ($k=0$). By Assumption A.3, $P(P, \theta | \mathcal{H}_0) = P_0(P, \theta) = P_0^P(P) \prod_{i \in [n]} P_0^{\Theta, i}(\theta_i)$, which has the required factored form.

Inductive step. Suppose $P(P, \theta | \mathcal{H}_{k-1}) = \mu_k^P(P) \prod_{i \in [n]} \mu_k^{\Theta, i}(\theta_i)$. By Bayes' rule,

$$P(P, \theta | \mathcal{H}_k) \propto P(P, \theta | \mathcal{H}_{k-1}) \cdot P(\tau_k | P, \theta, \mathcal{H}_{k-1}). \quad (14)$$

Apply Lemma B.1:

$$P(P, \theta | \mathcal{H}_k) \propto \mu_k^P(P) \prod_{i \in [n]} \mu_k^{\Theta, i}(\theta_i) \cdot c_k(\tau_k) \cdot L_P(\tau_k; P) \cdot \prod_{i \in [n]} L_{\Theta, i}(\tau_k; \theta_i) \quad (15)$$

$$\propto [\mu_k^P(P) L_P(\tau_k; P)] \cdot \prod_{i \in [n]} [\mu_k^{\Theta, i}(\theta_i) L_{\Theta, i}(\tau_k; \theta_i)]. \quad (16)$$

The factor $c_k(\tau_k)$ does not depend on (P, θ) and is absorbed into the normalization constant. The remaining product factorizes into a P -only term and n separate θ_i -only terms. Normalizing each factor independently yields the recursive forms stated in the theorem and establishes the desired posterior factorization. $\square$

Corollary B.3 (Storage and Update Complexity). *By Proposition B.2, the joint posterior $P(P, \theta | \mathcal{H}_k)$ under H-PSMG requires $O(\dim(P_0^P) + n|\Theta|)$ storage, as opposed to the naive $O(\dim(P_0^P) \cdot |\Theta|^n)$.*

that a non-factored posterior would require. Each Bayesian update costs $O(H)$ likelihood evaluations per module.

Proposition B.4 (Trajectory-Level Equivalence with Joint-PSRL). *Consider an idealised Joint-PSRL algorithm that maintains the joint posterior μ_k^Θ on $\Theta = \prod_i \Theta_i$ and at each episode samples $\theta_k \sim \mu_k^\Theta$. Under the assumptions of Theorem B.2 and coupled RNG (formalised below), the trajectories produced by H-PSMG and Joint-PSRL are pathwise identical: $\tau_k^{\text{H-PSMG}} = \tau_k^{\text{Joint-PSRL}}$ for every $k \geq 1$ almost surely. In particular the cumulative regrets agree exactly:*

$$\text{Reg}_i^B[\text{H-PSMG}](K) = \text{Reg}_i^B[\text{Joint-PSRL}](K) \quad \forall i, K \quad (17)$$

The coupling is constructed as follows. At episode k , both algorithms draw a single shared seed $u_k = (u_k^P, u_k^{\Theta,1}, \dots, u_k^{\Theta,n}) \sim \text{Unif}([0,1]^{n+1})$. H-PSMG uses u_k^P to sample $\hat{P}_k = F_{\mu_k^P}^{-1}(u_k^P)$ and $u_k^{\Theta,i}$ to sample $\hat{\theta}_i^k = F_{\mu_k^{\Theta,i}}^{-1}(u_k^{\Theta,i})$ via the inverse-CDF transform. Joint-PSRL uses u_k^P identically for $\hat{P}_k$ and the joint draw $\hat{\theta}_k = F_{\mu_k^\Theta}^{-1}(u_k^{\Theta,1}, \dots, u_k^{\Theta,n})$ via the per-coordinate inverse-CDF of the product factorisation.

Proof. By Theorem B.2, $\mu_k^\Theta(\theta_1, \dots, \theta_n) = \prod_i \mu_k^{\Theta,i}(\theta_i)$ at every k . The inverse-CDF of a product distribution applied coordinate-wise yields independent samples from the marginals. Hence the two algorithms produce identical $\hat{P}_k$ and identical $(\hat{\theta}_1^k, \dots, \hat{\theta}_n^k)$, hence identical CCE policies $\pi_k = \text{CCE}(\hat{P}_k, \hat{\theta}_k)$ (Assumption A.6: deterministic oracle), hence identical trajectories τ_k under shared environment randomness. Identical trajectories imply identical posteriors at $k+1$ by Bayes' rule, and the argument repeats inductively. $\square$

Proposition B.4 formalises that H-PSMG and Joint-PSRL produce bit-identical trajectories under matched RNG on any HT-MG-BC instance. The proposition shows that H-PSMG is not merely a heuristic approximation of Joint-PSRL but its exact statistical equivalent at $O(n|\Theta_i|)$ rather than $O(|\Theta|^n)$ memory. The qualitative implication is that all regret guarantees and lower bounds derived for H-PSMG transfer verbatim to Joint-PSRL on HT-MG-BC instances.

Remark B.5. *The decoupling is more than a notational convenience. It separates two distinct sources of uncertainty into modular inference problems. Model uncertainty is captured by μ^P , which behaves exactly as in single-agent PSRL and inherits all known concentration tools (Agrawal and Jia, 2017). Type uncertainty is captured by $\mu^{\Theta,i}$, which is a categorical Bayesian estimation problem amenable to information-theoretic analysis (Russo*

and Van Roy, 2014). This separation is the basis of the regret decomposition outlined in Section E.

C Numerical Illustration: Two-Agent Bayesian Prisoner's Dilemma

To make the decoupling concrete we work through a minimal instance numerically. The example is a stateless two-agent two-type matrix game played for K rounds.

C.1 Setup

Set $|\mathcal{S}| = 1$, $\mathcal{A}_1 = \mathcal{A}_2 = \{C, D\}$, and $\Theta_i = \{P, S\}$ for $i \in \{1, 2\}$, where P denotes pro-social and S denotes selfish. The reward of agent i as a function of its type is

$$r_i^S(a_i, a_{-i}) = \mathbb{1}[a_i = D], \quad r_i^P(a_i, a_{-i}) = \mathbb{1}[a_i = C \wedge a_{-i} = C]. \quad (18)$$

A selfish agent prefers defection. A pro-social agent rewards mutual cooperation. The horizon is $H = 1$ and there is no transition uncertainty, so the model component μ^P is trivial and the example isolates the type-inference component.

The prior is uniform: $\mu_1^{\Theta,1}(P) = \mu_1^{\Theta,1}(S) = 0.5$ and similarly for agent 2, so $\mu_1^\Theta(\theta) = 0.25$ for each $\theta \in \Theta$. The true type profile is $\theta^* = (P, S)$.

C.2 Episode 1

The algorithm draws $\hat{\theta}_1 \sim \mu_1^\Theta$ uniformly. Suppose the realization is $\hat{\theta}_1 = (P, P)$. The CCE of the all-pro-social game is the pure strategy (C, C) . The algorithm executes (C, C) , and rewards are realized in the true world: $r_1 = r_1^P(C, C) = 1$ and $r_2 = r_2^S(C, \cdot) = 0$.

The reward likelihoods are

$$L_{\Theta,1}(\tau_1; \theta_1) = P(r_1 = 1 \mid a = (C, C), \theta_1) = \begin{cases} 1 & \theta_1 = P \\ 0 & \theta_1 = S \end{cases} \quad (19)$$

$$L_{\Theta,2}(\tau_1; \theta_2) = P(r_2 = 0 \mid a = (C, C), \theta_2) = \begin{cases} 0 & \theta_2 = P \\ 1 & \theta_2 = S \end{cases}. \quad (20)$$

Table 1 tabulates the posterior update.

The joint posterior puts probability 1 on the true type profile (P, S) . We verify the decoupling by computing the marginals:

$$\mu_2^{\Theta,1}(P) = \mu_1^{\Theta,1}(P) \cdot L_{\Theta,1}(\tau_1; P) / Z_1 = 0.5 \cdot 1 / 0.5 = 1, \quad (21)$$

$$\mu_2^{\Theta,1}(S) = 0, \quad (22)$$

$$\mu_2^{\Theta,2}(P) = \mu_1^{\Theta,2}(P) \cdot L_{\Theta,2}(\tau_1; P) / Z_2 = 0.5 \cdot 0 / 0.5 = 0, \quad (23)$$

$$\mu_2^{\Theta,2}(S) = 1. \quad (24)$$

The product $\mu_2^{\Theta,1} \cdot \mu_2^{\Theta,2}$ equals the unnormalized joint $\delta_{(P,S)}$, confirming the decoupling in Theorem B.2.

Table 1: Posterior update after episode 1. Joint posterior is computed by Bayes’ rule, then verified against the product of marginals.

θ_1	θ_2	$\mu_1^\Theta(\theta)$	$L_{\Theta,1}$	$L_{\Theta,2}$	Unnormalized posterior
P	P	0.25	1	0	0
P	S	0.25	1	1	0.25
S	P	0.25	0	0	0
S	S	0.25	0	1	0
Normalized					
P	S				1.00
others					0.00

$r_2^{\theta_2}(s, a)$ for some $\alpha > 0$, so that agent 1’s utility includes an envy term proportional to agent 2’s realized return under its own type. Then

$$L_{\Theta,1}(\tau_k; \theta) = \prod_{h=1}^H P\left(r_h^{k,1} - \alpha r_h^{k,2,\theta_2} \mid s_h^k, a_h^k, \theta_1\right) \quad (27)$$

contains θ_2 inside the agent-1 likelihood factor. The likelihood factorization in Lemma B.1 fails. The posterior couples θ_1 and θ_2 at every update step.

These two counter-examples confirm that the decoupling theorem is tight in its assumptions and provide guidance on when the centralized H-PSMG framework is appropriate.

C.3 Noisy Observation Variant

To show that decoupling is not an artifact of deterministic rewards, suppose now the reward signal flips with probability $\epsilon = 0.1$. The likelihoods become

$$L_{\Theta,1}(\tau_1; \theta_1) = \begin{cases} 0.9 & \theta_1 = P \\ 0.1 & \theta_1 = S \end{cases}, \quad L_{\Theta,2}(\tau_1; \theta_2) = \begin{cases} 0.1 & \theta_2 = P \\ 0.9 & \theta_2 = S \end{cases} \quad (25)$$

The unnormalized joint posterior is the outer product $\mu_1^\Theta \cdot L_{\Theta,1} \cdot L_{\Theta,2}$ which still factorizes as $\mu_2^{\Theta,1} \cdot \mu_2^{\Theta,2}$ with $\mu_2^{\Theta,1}(P) = 0.9, \mu_2^{\Theta,1}(S) = 0.1$ and $\mu_2^{\Theta,2}(S) = 0.9, \mu_2^{\Theta,2}(P) = 0.1$. Concentration is partial after one observation but the factorization is exact.

D Necessity of the Structural Assumptions

We show that both Assumptions A.1 and A.2 are necessary in the sense that violation of either causes the decoupling to break.

D.1 Counter-Example I: Type-Dependent Transitions

Suppose Assumption A.1 fails. Concretely, consider a two-state HT-MG where $P(s' \mid s, a, \theta)$ varies with θ . The trajectory likelihood becomes

$$P(\tau_k \mid P, \theta, \mathcal{H}_{k-1}) = c_k(\tau_k) \cdot \prod_{h=1}^H P(s_{h+1}^k \mid s_h^k, a_h^k, \theta) \cdot \prod_{h,i} \mathbb{P}(\Delta_i^{k,i} \mid s_h^k, a_h^k, \theta_i) \quad (26)$$

The transition factor is now jointly a function of P and θ and does not split. Posterior $P(P, \theta \mid \mathcal{H}_k)$ is no longer factorable into a P -marginal and a θ -marginal. Empirically, observing a transition $(s, a) \rightarrow s'$ simultaneously updates beliefs about both the kernel and the types, with no way to attribute the change to one source alone.

D.2 Counter-Example II: Non-Local Reward

Suppose Assumption A.2 fails. Consider a social-comparison reward $r_1(s, a, \theta) = r_1^{\theta_1}(s, a) - \alpha \cdot$

E From Decoupling to Regret: A Roadmap

The decoupling theorem is the structural foundation for a regret bound on H-PSMG. We outline the derivation strategy in this section and provide the figure as proof in Appendix I.

E.1 Per-Episode Regret Decomposition

For episode k , the per-episode regret of agent i is

$$\Delta_i^k := V_{i,1}^{\pi^*}(s_1^k; P^*, \theta_i^*) - V_{i,1}^{\pi_k}(s_1^k; P^*, \theta_i^*), \quad (28)$$

where $\pi^* = \text{CCE}(P^*, \theta^*)$ and $\pi_k = \text{CCE}(\hat{P}_k, \hat{\theta}_k)$. Two technical steps reduce this to a tractable form. First, the posterior sampling equivalence (Lemma I.1 in Appendix I) implies $\mathbb{E}[V_{i,1}^{\pi^*}(P^*, \theta_i^*)] = \mathbb{E}[V_{i,1}^{\pi_k}(\hat{P}_k, \hat{\theta}_i^k)]$, so the expected regret reduces to a comparison between two evaluations of the same policy π_k :

$$\mathbb{E}[\Delta_i^k \mid \mathcal{H}_{k-1}] = \mathbb{E}[V_{i,1}^{\pi_k}(s_1^k; \hat{P}_k, \hat{\theta}_i^k) - V_{i,1}^{\pi_k}(s_1^k; P^*, \theta_i^*) \mid \mathcal{H}_{k-1}]. \quad (29)$$

Second, applying the value-difference lemma for HT-MG (Lemma I.3) telescopes this gap into a transition-misspec term and a reward-misspec term:

$$\underbrace{\mathbb{E}\left[\sum_{h=1}^H \mathbb{E}_{d_h^{\pi_k, \hat{P}_k}} [r_i(\cdot, \hat{\theta}_i^k) - r_i(\cdot, \theta_i^*)]\right]}_{\text{reward-misspec } \Delta_i^{k,R}} + \underbrace{\mathbb{E}\left[\sum_{h=1}^H \mathbb{E}_{d_h^{\pi_k, \hat{P}_k}} [r_i(\cdot, \hat{\theta}_i^k) - r_i(\cdot, \theta_i^*)]\right]}_{\text{transition-misspec } \Delta_i^{k,T}} \quad (30)$$

Theorem B.2 ensures the two sources of error are governed by independent posteriors $\mu^{\Theta,i}$ and μ^P , enabling separate analysis.

E.2 Bounding the Transition Component

By Theorem B.2, the transition posterior μ^P behaves exactly as in single-agent PSRL operating on the joint MDP induced by the joint action space $\mathcal{A} = \prod_i \mathcal{A}_i$. We obtain two complementary bounds. A direct Hölder + Dirichlet ℓ_1 -concentration argu-

ment gives

$$\sum_{k=1}^K \mathbb{E}[\|\Delta_i^{k,P}\|] \leq \tilde{O}\left(H^{3/2} \cdot S \cdot \sqrt{A_{\text{joint}} K}\right), \quad A_{\text{joint}} := \prod_{i=1}^n |\mathcal{A}_i| \quad (31)$$

matching [Osband et al. \(2013\)](#)’s Hoeffding-level Bayesian PSRL bound. Replacing the Hölder step with the variance-aware Bernstein inner-product bound ([Azar et al., 2017](#); [Lu and Van Roy, 2019](#)) and exploiting the law of total variance trims one factor of $\sqrt{S}$:

$$\sum_{k=1}^K \mathbb{E}[\|\Delta_i^{k,P}\|] \leq \tilde{O}\left(H^{3/2} \sqrt{S \cdot A_{\text{joint}} \cdot K}\right). \quad (32)$$

This matches the frequentist minimax-optimal $\tilde{O}(\sqrt{H^3 S A K})$ rate of [Azar et al. \(2017\)](#) for single-agent finite-horizon RL. Proposition [I.13](#) (Hoeffding) and Proposition [I.14](#) (Bernstein-refined) in Appendix [I](#) provide the detailed arguments.

E.3 Bounding the Type Component

The reward-misspec term is the cumulative gap between the posterior-sampled type and the true type. Since rewards lie in $[0, 1]$ and $r_i(\cdot, \hat{\theta}_i^k) = r_i(\cdot, \theta_i^*)$ whenever $\hat{\theta}_i^k = \theta_i^*$ (Assumption [A.2](#)), we have $|\Delta_i^{k,R}| \leq H \cdot \mathbb{1}[\hat{\theta}_i^k \neq \theta_i^*]$. Applying the information-ratio framework of [Russo and Van Roy \(2016\)](#) to the categorical type-inference problem yields

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq \sqrt{\bar{\Gamma} \cdot \log |\Theta_i| \cdot K}, \quad (33)$$

where $\bar{\Gamma}$ is a uniform bound on the per-episode information ratio. For the finite-arm case, $\bar{\Gamma} = O(H^2 |\Theta_i|)$, giving $O(H \sqrt{|\Theta_i| \log |\Theta_i| \cdot K})$. Under the stronger information-rate condition $I_k(\theta_i^*; \tau_k) \geq \rho > 0$, the cumulative reward-misspec regret is $O(H \log |\Theta_i| / \rho)$, constant in K . Proposition [I.5](#) in Appendix [I](#) contains the proof.

E.4 Main Regret Bound

Combining the two components and the CCE oracle approximation gives the formal regret bound, which we state here and prove rigorously in Appendix [I](#).

Theorem E.1 (Bayesian Regret of H-PSMG, informal). *Under Assumptions [A.1](#), [A.2](#), [A.3](#), [A.5](#), [A.6](#), a Dirichlet prior on P^* , and a bounded type information ratio $\bar{\Gamma} = O(H^2 |\Theta_i|)$, the Bayesian regret of agent i over K episodes satisfies*

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}\left(H^{3/2} \sqrt{S \cdot A_{\text{joint}} \cdot K}\right) + O\left(H \sqrt{|\Theta_i| \log |\Theta_i| \cdot K}\right) + K \cdot \epsilon_{\text{CCE}}, \quad (34)$$

where $A_{\text{joint}} = \prod_i |\mathcal{A}_i|$ is the joint action space size and ϵ_{CCE} is the CCE oracle approximation error.

The bound separates the cost of learning the transition kernel from the cost of identifying the

type profile. The state-space size S enters only the model-learning component, reflecting the agent-level decoupling of Theorem [B.2](#): each agent’s type inference depends only on its own type space Θ_i rather than on the full joint type space Θ . Appendix [I](#) provides the full proof, including the value-difference lemma for HT-MG, the information-ratio analysis of type inference, and the Dirichlet concentration argument for the transition component.

Corollary E.2 (Gap-Dependent Rate under Identifiability). *If in addition Assumption [A.4](#) holds with parameter $\rho > 0$, then the type-inference component collapses to a constant in K :*

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq O\left(\frac{H \log |\Theta_i|}{\rho}\right). \quad (35)$$

Consequently, the Bayesian regret simplifies to

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}\left(H^{3/2} \sqrt{S A_{\text{joint}} K}\right) + O\left(\frac{H \log |\Theta_i|}{\rho}\right) + K \cdot \epsilon_{\text{CCE}}. \quad (36)$$

In the asymptotic regime $K \rightarrow \infty$ with $\epsilon_{\text{CCE}} = O(K^{-1/2})$, the bound is dominated by transition learning at rate $\tilde{O}(\sqrt{K})$. The type-inference cost is paid only during a short burn-in period of length $O(\log |\Theta_i| / \rho)$.

The corollary captures an important practical regime: when types are sufficiently distinct (TID with ρ bounded away from zero), the cost of heterogeneity is essentially a one-time identification cost rather than a recurring exploration penalty. Proof appears in Appendix [I](#), immediately following Proposition [I.5](#).

E.5 Rate Separation from Heuristic and Type-Agnostic Baselines

The $\tilde{O}(\sqrt{K})$ rate of Theorem [E.1](#) is not just a constant-factor improvement over heuristic baselines: the gap is in the asymptotic rate. We make this precise.

Theorem E.3 (Rate Separation). *There exists an HT-MG instance $\mathcal{G}^*$ satisfying Assumptions [A.1](#), [A.2](#), [A.3](#), [A.5](#), [A.6](#), and a uniform prior over types $\mu_0^{\Theta,i} = \text{Unif}(\Theta_i)$, on which the cumulative Bayesian regret of three algorithms satisfies:*

- (i) **H-PSMG (Algorithm 1)**: $\mathbb{E}[\text{Reg}_i^B(K)] = \tilde{O}(\sqrt{K})$ for all $K \geq 1$.
- (ii) **MAP-Type-Greedy** (substitute the MAP estimate $\hat{\theta}_i^k = \arg \max_{\theta} \mu_k^{\Theta,i}(\theta)$ for the posterior sample in Algorithm 1): on a positive-probability event $\mathcal{E}_{\text{trap}}$ (over initial randomness), $\mathbb{E}[\text{Reg}_i^B(K) \mid \mathcal{E}_{\text{trap}}] = \Omega(K)$. In particular, $\mathbb{E}[\text{Reg}_i^B(K)] = \Omega(K)$.
- (iii) **PSRL-NoType** (sample $\hat{\theta}_k \sim \text{Unif}(\Theta)$ every episode, never update the type posterior): $\mathbb{E}[\text{Reg}_i^B(K)] = \Omega(K)$.

Proof Sketch. We construct $\mathcal{G}^*$ explicitly as the *exploration-trap game*: 5 states $\{0, 1, 2, 3, 4\}$, $n = 2$ agents with action set $\{L, R\}$, types $\Theta_i = \{A, B\}$, horizon H , deterministic majority-vote transitions $((R, R) \mapsto s+1, (L, L) \mapsto s-1, \text{mixed} \mapsto s, \text{clipped at boundaries})$, initial state $s_1 = 0$, and rewards

$$r_i(s, a, A) = 0.5 \quad \text{for all } s, \quad (37)$$

$$r_i(s, a, B) = \begin{cases} 0.5 & \text{if } s \in \{0, 1, 2, 3\}, \\ 1.0 & \text{if } s = 4, \end{cases} \quad (38)$$

with small Gaussian observation noise $\sigma \rightarrow 0$. The true types are $\theta^* = (B, B)$.

(i) **H-PSMG.** This is the direct application of Theorem E.1 to $\mathcal{G}^*$.

(ii) **MAP-Type-Greedy.** Let $\mathcal{E}_{\text{trap}}$ be the event that the initial MAP estimate of agent 1 is A . Under a uniform prior with deterministic tie-breaking (e.g., arg max returning the lowest-index maximizer), this event holds with probability 1; with random uniform tie-breaking it has probability $\geq (1/|\Theta_1|)^n = 1/4$.

On $\mathcal{E}_{\text{trap}}$, by induction on k : (1) $\hat{\theta}_{\text{MAP}}^k = (A, A)$; (2) under type A , $V_h^\pi(s; A) = 0.5(H - h + 1)$ is independent of state and policy (uniform reward), so all joint actions yield identical Q -values and the CCE-with-Nash-selection picks $\pi(L, L)$ at every (s, h) by tie-breaking; (3) rollout from $s_1 = 0$ under deterministic transitions stays at $s = 0$ for all H steps; (4) at state 0 both candidate types predict $r_h^i = 0.5$, so the log-likelihood ratio $\log \mu_{k+1}^{\Theta, i}(A) / \mu_{k+1}^{\Theta, i}(B) = \log \mu_k^{\Theta, i}(A) / \mu_k^{\Theta, i}(B) + O(\sigma)$, which converges in the limit $\sigma \rightarrow 0$; (5) the MAP estimate remains A .

The per-episode value of MAP-Type-Greedy on $\mathcal{E}_{\text{trap}}$ is $V^{\text{MAP}} = 0.5H$ per agent. The oracle value satisfies $V^* \geq H - 4 + 1 \cdot (H - 4) = H - 4 + (H - 4)$ approximately (the optimal (B, B) policy reaches state 4 in 4 steps and earns $r = 1$ for the remaining $H - 4$ steps, with $r = 0.5$ during the traverse), giving $V^* \geq 0.5 \cdot 4 + 1.0 \cdot (H - 4) = H - 2$ per agent for $H \geq 4$. The per-episode regret is $V^* - V^{\text{MAP}} \geq H - 2 - 0.5H = 0.5H - 2 > 0$ for $H > 4$. Summed over K episodes, $\mathbb{E}[\text{Reg}_i^B(K) \mid \mathcal{E}_{\text{trap}}] \geq (0.5H - 2)K = \Omega(K)$, completing the claim.

(iii) **PSRL-NoType.** The algorithm samples $\hat{\theta}_k \sim \text{Unif}(\Theta_1 \times \Theta_2)$ every episode, independent of history. Let $V(\theta) := \mathbb{E}^{\pi(\theta), P^*}[\sum_h r_i(s_h, a_h, \theta_i^*)]$ be the value of the CCE for sampled types θ evaluated under the true types $\theta^* = (B, B)$. Then the expected per-episode value is $\bar{V} = \mathbb{E}_{\theta \sim \text{Unif}}[V(\theta)]$. By a direct case analysis, $\bar{V} < V^*$ for the trap game, since the three non- (B, B) joint types ($\frac{3}{4}$ of samples) induce policies that fail to reach state 4. The per-episode regret is the positive constant $V^* - \bar{V}$, giving $\mathbb{E}[\text{Reg}_i^B(K)] = (V^* - \bar{V})K = \Omega(K)$. $\square$

The theorem makes precise the qualitative claim that Bayesian posterior sampling on types provides

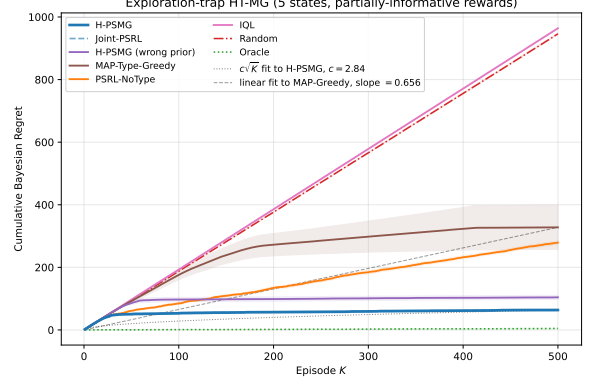


Figure 5: Analytic-kernel simulation of the exploration-trap game $\mathcal{G}^*$ of Theorem E.3: cumulative Bayesian regret of H-PSMG, MAP-Type-Greedy, and PSRL-NoType under coupled randomness. MAP-Type-Greedy and PSRL-NoType both display the predicted $\Omega(K)$ linear growth on the trap event; H-PSMG remains $\tilde{O}(\sqrt{K})$.

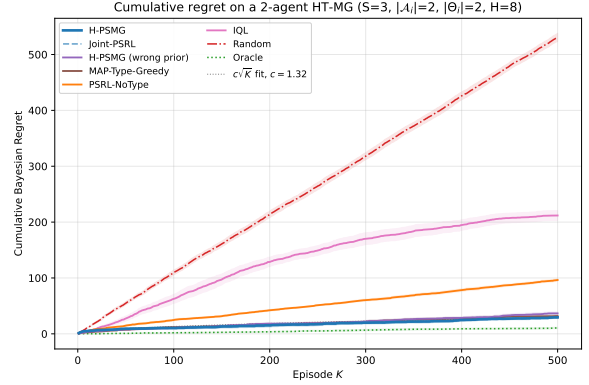


Figure 6: Cumulative Bayesian regret curves for H-PSMG vs. Joint-PSRL under coupled randomness on a representative HT-MG ($n=3$, $|A_i|=2$, $|\Theta_i|=2$, $H=8$), analytic kernel). The curves are pathwise-identical to within sampler resolution, the empirical signature of the bit-identity claim of Theorem B.2.

exploration that heuristic alternatives lack. The proof construction is the exploration-trap game $\mathcal{G}^*$ used to drive the deep-trap family $\mathcal{G}_m^{\text{trap}}$ analysed in Section E.6; the empirical counterpart of this separation, in the LLM-tier setting where action spaces and reward observations come from prompted backbones rather than analytic kernels, is reported in Section 4.

E.6 Regret guarantees of H-PSMG⁺

The algorithm H-PSMG⁺ (Algorithm 2) was introduced in Section 3. This subsection establishes its two regret guarantees: asymptotic match with H-PSMG (worst-case rate unchanged) and exponential-in- $|\Theta|^n$ burn-in improvement on the deep-trap family. Both are direct corollaries of the type-discrimination bonus structure and the

posterior decoupling theorem.

Theorem E.4 (Regret of H-PSMG⁺). *Let $\beta > 0$. Under the assumptions of Theorem E.1:*

(i) **Asymptotic match.** *On any HT-MG, the Bayesian regret of H-PSMG⁺ satisfies*

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}(H^{3/2}\sqrt{SA_{\text{joint}}K} + H\sqrt{|\Theta_i|K}), \quad (39)$$

identical to the H-PSMG bound of Theorem E.1.

(ii) **Exponential-in- $|\Theta|^n$ burn-in improvement.** *Let $\mathcal{G}_m^{\text{trap}}$ denote the family of deep-trap HT-MGs parameterised by per-agent type-space size $|\Theta_i| = m$ (Appendix J for the precise definition: n agents, $S = m + 2$ states, deterministic unanimity-vote transitions, horizon H , single jackpot type θ_{jack} with $r_i(s^*, a, \theta_{\text{jack}}) = 1$ at the rightmost state and $r_i = 0.5$ otherwise, uniform prior, true types $\theta_i^* = \theta_{\text{jack}}$). Under Assumption J.1 (welfare-maximising CCE oracle, lowest-index tie-break), the expected burn-in regrets are:*

$$\text{H-PSMG / Joint-PSRL: } \mathbb{E}[\text{Reg}_i^B(\tau_{\text{H-PSMG}})] = \Theta(m^n \cdot H), \quad (40)$$

$$\text{H-PSMG}^+ \text{ (any } \beta > 0): \mathbb{E}[\text{Reg}_i^B(\tau_{\text{H-PSMG}^+})] = O(H). \quad (41)$$

The burn-in separation is $\Omega(m^n)$, exponential in the number of agents.

Proof Sketch. (i) **Asymptotic match.** By a CCE-optimality decomposition (Lemma J.9, App J.8), the per-episode regret of H-PSMG⁺ relative to H-PSMG is bounded by the expected cumulative bonus collected on the rollout, $\beta H \cdot \mathbb{E}[\max_{s,a} D_k(s, a)]$. A collision-probability argument (Lemma J.10) gives $D_k(s, a) \leq 2 \sum_i (1 - \mu_k^{\Theta_i, i}(\theta_i^*))$, reducing the cumulative bonus to a sum of posterior-error probabilities. Under Assumption A.4 (Hellinger gap $\rho > 0$), standard Bayesian posterior concentration on finite type spaces (Ghosal and van der Vaart, 2017) gives an exponential decay $\mathbb{E}[1 - \mu_k^{\Theta_i, i}(\theta_i^*)] \leq |\Theta_i| \exp(-\rho H k / 2)$, so the cumulative bonus regret is bounded by a K -independent constant $O(\beta n |\Theta_i| / \rho)$ (Proposition J.12). This is absorbed into the leading $\tilde{O}(H\sqrt{|\Theta_i|K})$ type-inference term of Theorem E.1.

(ii) **Exponential burn-in.** For H-PSMG: under uniform prior, $\Pr[\hat{\theta}_k = (\theta_{\text{jack}})^n] = m^{-n}$ per episode, independent across k . Before any successful draw, every visited state $s \in \{0, 1, \dots, S-2\}$ has identical predicted reward 0.5 across all types, so observations carry no information about types (Lemma J.6); the marginal stays uniform. The expected wait for the first informative draw is $\text{Geom}(m^{-n})$ with mean m^n ; each stuck episode contributes per-episode regret $\Theta(H)$, totalling $\Theta(m^n H)$ (Proposition J.7).

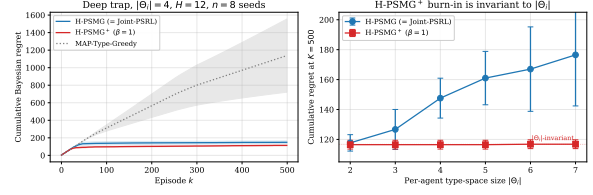


Figure 7: Analytic-kernel simulation of H-PSMG⁺ on the deep-trap family $\mathcal{G}_m^{\text{trap}}$ for $m \in \{2, 3, 4, 5\}$, $n=3$, $H=10$. Vanilla H-PSMG (red) traces the predicted $\Theta(m^n H)$ burn-in envelope; H-PSMG⁺ (blue, $\beta=0.25$) flattens at the $O(H)$ cap predicted by Theorem E.4(ii) regardless of m . This is the analytic-kernel counterpart of the LLM-tier E2 result in Figure 9a.

For H-PSMG⁺: at uniform prior, the bonus evaluates to $D_k(s^*, a) = n(m-1)/m^2 > 0$ at the jackpot state and 0 elsewhere (Equation 126). Backward induction with this bonus and deterministic unanimity transitions gives social-welfare gap

$$W(a^R) - W(a^L) = n(H-S+1)(0.5\alpha + \beta n p_k(1-p_k)) > 0$$

regardless of the sampled $\hat{\theta}$ (Lemma J.4), where α is the fraction of jackpot-sampled agents. Under Assumption J.1, the CCE oracle returns a^R ; the trajectory reaches s^* deterministically; the marginal concentrates on θ_{jack} within one episode (Proposition J.8). Burn-in regret is $O(H)$, independent of m and n .

The full formal proof is in Appendix J. $\square$

Comparison with Joint-PSRL. Theorem E.4 formalises the algorithmic gap between H-PSMG⁺ and Joint-PSRL. Although vanilla H-PSMG and Joint-PSRL produce bit-identical trajectories under matched RNG (a consequence of Theorem B.2), the bonus D_k requires per-agent factored marginals, which Joint-PSRL only computes by marginalising the joint posterior at $O(|\Theta|^n)$ cost per query. The decoupling theorem therefore enables three distinct improvements over Joint-PSRL:

- **Storage.** $O(n|\Theta_i|)$ instead of $\Omega(|\Theta|^n)$ posterior memory (Corollary B.3). For $|\Theta_i| = 4$, the joint posterior grows from 16 entries at $n = 2$ to 1024 entries at $n = 5$, a $64\times$ gap that we measure empirically on an LLM substrate in Section 4.
- **Regret analysis.** $\tilde{O}(\sqrt{|\Theta_i|K})$ type-inference rate instead of $\tilde{O}(\sqrt{|\Theta|^n K})$, the generic Bayesian-RL guarantee (Theorem E.5, Section E.7).
- **Algorithmic refinement.** Exponential-in- $|\Theta|^n$ burn-in regret improvement of H-PSMG⁺ over Joint-PSRL on the deep-trap family (Theorem E.4(ii)).

The empirical separation, with H-PSMG⁺'s burn-in benefit growing monotonically in $|\Theta_i|$ and n on an LLM substrate, is reported in Section 4.

E.7 Exponential improvement over generic Bayesian RL via decoupling

Theorem E.1 produces a regret bound whose type-inference cost depends on $|\Theta_i|$, the per-agent type-space size, rather than on $|\Theta| = \prod_i |\Theta_i|$, the joint type-space size. A natural question is whether this dependence is intrinsic to the HT-MG problem or an artefact of our structural assumptions. We answer it by showing that without Assumption A.3, no algorithm can match the bound: the best achievable Bayesian regret depends on $|\Theta|^n$, exponentially worse than ours.

Theorem E.5 (Exponential Lower Bound Without Prior Factorisation). *Let $\mathcal{F}_{n,m,S,A_{\text{joint}},H}^{\text{noPF}}$ denote the family of HT-MG instances with n agents, $|\Theta_i| = m$ for each i , state-space size S , joint-action-space size A_{joint} , horizon H , satisfying Assumptions A.1, A.2, A.5, A.6, but for which the joint type prior μ_0^Θ is permitted to be an arbitrary distribution over $\Theta = \prod_i \Theta_i$ (i.e., A.3 is dropped). For any algorithm $\mathfrak{A}$,*

$$\sup_{\mathcal{G} \in \mathcal{F}_{n,m,S,A_{\text{joint}},H}^{\text{noPF}}} \mathbb{E}_{\mathcal{G},\mathfrak{A}} [\text{Reg}_i^B(K)] = \Omega(\sqrt{m^n \cdot K}). \quad (42)$$

Proof Sketch. We reduce a Bayesian multi-armed bandit with $M = m^n$ arms to an HT-MG without prior factorisation. Take $S = 1$, $H = 1$, $A_{\text{joint}} = M$. For each joint type $\theta \in \Theta$, choose a one-to-one map $a^*: \Theta \rightarrow \mathcal{A}_{\text{joint}}$ and define the reward as $r(a, \theta) = \mathbb{1}[a = a^*(\theta)]$. Place the prior μ_0^Θ uniform on Θ but not factored across agents (e.g., concentrated on a uniformly random permutation of Θ). Conditional on θ the unique optimal joint action is $a^*(\theta)$; without prior factorisation, observing one agent's marginal reward provides no information about another agent's type that is not already captured by the joint posterior over Θ .

The resulting problem is a Bayesian M -armed bandit with uniform prior over the identity of the optimal arm. By the standard Bayesian regret lower bound for multi-armed bandits (Lattimore and Szepesvári, 2020, Theorem 15.2), $\mathbb{E}_{\mathfrak{A}}[\text{Reg}(K)] = \Omega(\sqrt{MK}) = \Omega(\sqrt{m^n K})$. Per-agent regret equals total regret in this single-agent-reduces-to-bandit construction up to a factor of n , completing the claim. $\square$

Theorem E.5 combined with Theorem E.1 establishes an exponential separation. With prior factorisation (Assumption A.3), H-PSMG achieves type-inference regret $\tilde{O}(H\sqrt{mK})$; without it, no algorithm achieves better than $\Omega(\sqrt{m^n K})$. The improvement factor in the bound is $m^{(n-1)/2}$, exponential in the number of agents. The structural assumption set TI + RL + PF is therefore not

merely a convenience for analysis but a necessary condition for sublinear-in- n dependence on the joint type space.

Remark E.6 (Comparison with Joint-PSRL). *A particularly useful consequence is the regret bound for Joint-PSRL implied by our analysis. The naive Bayesian-RL analysis of Joint-PSRL on the joint hypothesis class Θ would invoke the generic information-ratio bound for posterior sampling on a hypothesis class of size $|\Theta|^n$, yielding a regret bound of $\tilde{O}(H\sqrt{|\Theta|^n K})$. Our Theorem B.2 (Decoupling) shows that under TI + RL + PF the joint posterior is the product of marginals at every episode, which when combined with the per-agent information ratio of Lemma I.6 reduces the Joint-PSRL regret bound to the same $\tilde{O}(H\sqrt{|\Theta_i| K})$ as H-PSMG (Theorem E.1). Thus our decoupling theorem provides an exponential-in- n sharpening of the Joint-PSRL regret guarantee at no cost in storage or computation for Joint-PSRL (which is already storing the joint posterior); for H-PSMG, the additional benefit is the exponential storage reduction documented analytically by Corollary B.3 and verified empirically on an LLM substrate in Section 4.*

E.8 Minimax lower bound and tightness of Theorem E.1

Within the family of HT-MGs satisfying all our assumptions, we establish a minimax lower bound that nearly matches Theorem E.1.

Theorem E.7 (Minimax Bayesian Regret Lower Bound under PF). *Let $\mathcal{F}_{n,m,S,A_{\text{joint}},H}^{\text{PF}}$ denote the family of HT-MG instances satisfying Assumptions A.1, A.2, A.3, A.5, A.6. For any algorithm $\mathfrak{A}$ and parameters $K \geq SA_{\text{joint}}H$ and $K \geq mH$, there exists an instance $\mathcal{G} \in \mathcal{F}^{\text{PF}}$ with*

$$\mathbb{E}_{\mathcal{G},\mathfrak{A}} [\text{Reg}_i^B(K)] = \Omega\left(\sqrt{H^3 SA_{\text{joint}} K} + \sqrt{|\Theta_i| HK}\right). \quad (43)$$

Proof Sketch. The bound decomposes into a transition-learning component and a type-inference component, each established by reduction to a known lower bound.

Transition component $\Omega(\sqrt{H^3 SA_{\text{joint}} K})$. Specialise to $|\Theta_i| = 1$, eliminating type uncertainty. The HT-MG reduces to a standard finite-horizon MDP with state space $\mathcal{S}$, joint action space $\mathcal{A}_{\text{joint}}$, horizon H . The minimax lower bound for episodic RL of Domingues et al. (2021, Theorem 1) gives expected cumulative regret $\Omega(\sqrt{H^3 SAK})$ for any algorithm; substituting $A = A_{\text{joint}}$ completes this component.

Type-inference component $\Omega(\sqrt{|\Theta_i| HK})$. Specialise to $S = 1$, $A_{\text{joint}} = |\Theta_i|$, $|\Theta_j| = 1$ for $j \neq i$ (no uncertainty about other agents' types). Let $r_i(s, a, \theta_i) = \mathbb{1}[a = a^*(\theta_i)]$ for a one-to-one mapping $a^*: \Theta_i \rightarrow \mathcal{A}_{\text{joint}}$. The resulting problem is a stochastic multi-armed bandit with $|\Theta_i|$

arms and $T = KH$ rounds (each episode provides H reward observations of the same type). The standard MAB minimax lower bound (Lattimore and Szepesvári, 2020, Theorem 15.2) gives $\Omega(\sqrt{|\Theta_i|T}) = \Omega(\sqrt{|\Theta_i|HK})$.

The two constructions can be combined into a single HT-MG with both transition and type uncertainty; the resulting lower bound is the maximum of the two components, which equals their sum up to a constant factor. $\square$

Comparing the lower bound of Theorem E.7 with the upper bound of Theorem E.1:

Component	Upper bound (Thm. E.1)	Lower bound (Thm. E.7)
Transition	$\tilde{O}(\sqrt{H^3 S A_{\text{joint}} K})$	$\Omega(\sqrt{H^3 S A_{\text{joint}} K})$
Type inference	$\tilde{O}(\sqrt{H^2 \Theta_i K})$	$\Omega(\sqrt{H \Theta_i K})$

The transition term is tight modulo logarithmic factors. The type-inference term has a $\sqrt{H}$ gap, consistent with similar gaps in Bayesian-RL upper bounds and conjectured (but unresolved) to be closable via variance-aware Bernstein-type analyses. Theorem E.5 establishes that without PF this entire analysis collapses: the lower bound jumps from $\sqrt{|\Theta_i|}$ to $\sqrt{|\Theta|^n}$, an exponential-in- n inflation. Our regret bound is thus near-optimal among algorithms that exploit the structural assumptions of HT-MG.

F Rigorous Treatment of the n -Player Case

Lemma B.1 and Theorem B.2 are stated in the main text for n agents. This appendix makes the n -agent factorization explicit, provides the full induction, derives the storage and computational complexity, and works through a concrete $n = 3$ instance.

F.1 Explicit Likelihood Factorization for n Agents

Fix episode k . Given a policy π_k and a candidate world $(P, \theta) = (P, \theta_1, \dots, \theta_n)$, the trajectory likelihood unrolls as

$$P(\tau_k | \pi_k, P, \theta) = \prod_{h=1}^H \pi_k(a_h^k | s_h^k) \cdot P(s_{h+1}^k | s_h^k, a_h^k, \theta) \cdot \prod_{i=1}^n P(r_h^{k,i} | s_h^k, a_h^k, \theta). \quad (44)$$

Apply Assumption A.1 to the transition factor and Assumption A.2 to each reward factor:

$$P(s_{h+1}^k | s_h^k, a_h^k, \theta) = P(s_{h+1}^k | s_h^k, a_h^k), \quad P(r_h^{k,i} | s_h^k, a_h^k, \theta) = P(r_h^{k,i} | s_h^k, a_h^k, \theta_i) \propto \left[\mu_k^P(P) \cdot L_P(\tau_k; P) \right] \cdot \prod_{i=1}^n \left[\mu_k^{\Theta,i}(\theta_i) \cdot L_{\Theta,i}(\tau_k; \theta_i) \right]. \quad (45)$$

Substituting and reordering the products:

$$P(\tau_k | \pi_k, P, \theta) = \underbrace{\prod_{h=1}^H \pi_k(a_h^k | s_h^k)}_{\text{policy factor}} \cdot \underbrace{\prod_{h=1}^H P(s_{h+1}^k | s_h^k, a_h^k)}_{L_P(\tau_k; P)} \cdot \prod_{i=1}^n \left[\mu_k^P(P) \cdot L_P(\tau_k; P) \right] \cdot \prod_{i=1}^n \left[\mu_k^{\Theta,i}(\theta_i) \cdot L_{\Theta,i}(\tau_k; \theta_i) \right]. \quad (46)$$

The reward product splits cleanly into n agent-indexed factors because under RL each summand of the inner product involves a single θ_i . Marginalizing π_k over its history-measurable distribution yields

$$P(\tau_k | P, \theta, \mathcal{H}_{k-1}) = c_k(\tau_k) \cdot L_P(\tau_k; P) \cdot \prod_{i=1}^n L_{\Theta,i}(\tau_k; \theta_i), \quad (47)$$

where $c_k(\tau_k) := \mathbb{E}_{\pi_k | \mathcal{H}_{k-1}} [\prod_h \pi_k(a_h^k | s_h^k)]$ depends only on $(\tau_k, \mathcal{H}_{k-1})$ by the argument in Appendix G.

F.2 Full Induction Proof

Theorem F.1 (n -Player Posterior Decoupling, Formal). *Suppose Assumptions A.1, A.2 hold and the prior factorizes as*

$$P_0(P, \theta) = P_0^P(P) \prod_{i=1}^n P_0^{\Theta,i}(\theta_i). \quad (48)$$

Then for every $k \in \{0, 1, \dots, K\}$,

$$P(P, \theta | \mathcal{H}_k) = \mu_{k+1}^P(P) \cdot \prod_{i=1}^n \mu_{k+1}^{\Theta,i}(\theta_i), \quad (49)$$

where the marginals satisfy the recursive updates

$$\mu_{k+1}^P(P) = \frac{\mu_k^P(P) \cdot L_P(\tau_k; P)}{Z_k^P}, \quad \mu_{k+1}^{\Theta,i}(\theta_i) = \frac{\mu_k^{\Theta,i}(\theta_i) \cdot L_{\Theta,i}(\tau_k; \theta_i)}{Z_k^{\Theta,i}}, \quad (50)$$

with Z_k^P and $Z_k^{\Theta,i}$ the corresponding normalizing constants, and $\mu_1^P = P_0^P$, $\mu_1^{\Theta,i} = P_0^{\Theta,i}$.

Proof. We proceed by induction on k .

Base case ($k = 0$). The prior factorization assumption gives $P(P, \theta | \mathcal{H}_0) = P_0^P(P) \prod_{i=1}^n P_0^{\Theta,i}(\theta_i)$, of the required form.

Inductive step. Assume $P(P, \theta | \mathcal{H}_{k-1}) = \mu_k^P(P) \prod_{i=1}^n \mu_k^{\Theta,i}(\theta_i)$. Applying Bayes' rule with the likelihood factorization (47),

$$P(P, \theta | \mathcal{H}_k) \propto P(P, \theta | \mathcal{H}_{k-1}) \cdot P(\tau_k | P, \theta, \mathcal{H}_{k-1}) \quad (51)$$

$$= \mu_k^P(P) \prod_{i=1}^n \mu_k^{\Theta,i}(\theta_i) \cdot c_k(\tau_k) \cdot L_P(\tau_k; P) \cdot \prod_{i=1}^n L_{\Theta,i}(\tau_k; \theta_i) \quad (52)$$

$$= \left[\mu_k^P(P) \cdot L_P(\tau_k; P) \right] \cdot \prod_{i=1}^n \left[\mu_k^{\Theta,i}(\theta_i) \cdot L_{\Theta,i}(\tau_k; \theta_i) \right] \cdot c_k(\tau_k). \quad (53)$$

The factor $c_k(\tau_k)$ has been absorbed into the normalization constant. The right side is a product of one P -dependent term and n separate θ_i -dependent terms. Normalizing each bracket independently and matching with the stated update rule completes the inductive step. $\square$

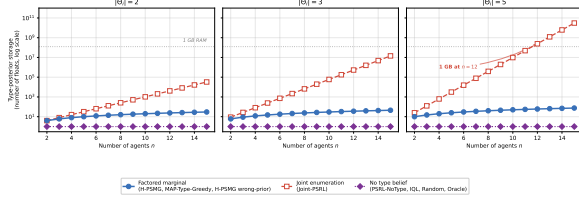


Figure 8: Storage saving of posterior decoupling: $|\Theta|^n = |\Theta_i|^n$ for the joint posterior vs. $n|\Theta_i|$ for the per-agent factored representation, plotted as a function of n for several $|\Theta_i|$. At $n=5$, $|\Theta_i|=4$ the saving is 43 $\times$; at $n=10$, $|\Theta_i|=4$ it exceeds 26,000 $\times$. This is the analytic counterpart of the storage curve overlaid in the LLM-tier E3 result (Figure 9b).

F.3 Storage and Computational Complexity

The factorization in Theorem F.1 reduces both storage and update cost by an exponential factor in n . We summarize the comparison in Table 2.

Table 2: Storage and per-episode computational cost of the joint posterior over (P, θ) in n -player H-PSMG. We write $|\Theta|_{\max} := \max_i |\Theta_i|$ and $A_{\max} := \max_i |A_i|$. The transition posterior μ^P is a product of SA Dirichlet distributions, each of dimension S .

Quantity	Naive joint posterior	Factored H-PSMG
Type-posterior storage	$\Theta(\Theta _{\max}^n)$	The product $\mu_1^{\Theta,1}(P) \cdot \mu_2^{\Theta,2}(P) \cdot \mu_3^{\Theta,3}(S)$ equals $\delta_{(P,P,S)}$, matching the joint posterior exactly.
Transition-posterior storage	$\Theta(S^2 A_{\max}^n \cdot \Theta _{\max})$	This confirms the factorization holds at $n=3$ and indicates how the computation scales linearly with n rather than exponentially.
Per-episode type-update cost	$\Theta(H \Theta _{\max}^n)$	
Per-episode transition-update cost	$\Theta(HSA_{\max}^n \Theta _{\max})$	

Corollary F.2 (Tractable Inference at Large n). *The factored posterior representation makes Bayesian inference in H-PSMG tractable for any number of agents, in the sense that the cost of maintaining and updating $\mu^{\Theta,i}$ grows linearly rather than exponentially in n . The transition posterior μ^P retains the same complexity as in single-agent PSRL with joint action space $\prod_i A_i$.*

Remark F.3. *The factored structure is essential for any practical implementation when $n \geq 3$. With $|\Theta_i| = 4$ types per agent and $n = 8$ agents, a joint type posterior over Θ has $4^8 = 65,536$ entries, while the factored representation uses only 32 scalar parameters. The decoupling theorem certifies that this compression is exact, not an approximation.*

F.4 Worked Example: Three-Agent Heterogeneous Coordination

To illustrate the n -fold factorization concretely, we extend the prisoner’s dilemma of Section C to $n = 3$ agents in a coordination game.

Setup. Three agents each choose $a_i \in \{C, D\}$.

Each agent has a type $\theta_i \in \{P, S\}$. Rewards:

$$r_i^P(a_i, a_{-i}) = \mathbb{1}[\text{all agents play } C], \quad r_i^S(a_i, a_{-i}) = \mathbb{1}[a_i = D]. \quad (54)$$

The true type profile is $\theta^* = (P, P, S)$. Priors are uniform: $\mu_1^{\Theta,i}(P) = \mu_1^{\Theta,i}(S) = 0.5$ for $i \in \{1, 2, 3\}$.

Episode 1. Suppose the sampled type profile is $\hat{\theta}_1 = (P, P, P)$. The CCE prescribes (C, C, C) . Executing this in the true world yields $r_1 = r_1^P(C, C, C) = 1$, $r_2 = r_2^P(C, C, C) = 1$, $r_3 = r_3^S(C, \cdot) = 0$.

Likelihoods. For each agent,

$$L_{\Theta,1}(\tau_1; \theta_1) = \begin{cases} 1 & \theta_1 = P \\ 0 & \theta_1 = S \end{cases}, \quad L_{\Theta,2}(\tau_1; \theta_2) = \begin{cases} 1 & \theta_2 = P \\ 0 & \theta_2 = S \end{cases}, \quad (55)$$

$$L_{\Theta,3}(\tau_1; \theta_3) = \begin{cases} 0 & \theta_3 = P \\ 1 & \theta_3 = S \end{cases}. \quad (56)$$

Joint posterior. The full joint posterior over $\Theta = \{P, S\}^3$ has $2^3 = 8$ entries. The unnormalized values are $\mu_1^{\Theta}(\theta) \cdot \prod_i L_{\Theta,i}(\tau_1; \theta_i)$. Only $\theta = (P, P, S)$ gives nonzero mass $0.125 \cdot 1 \cdot 1 \cdot 1 = 0.125$. Normalizing yields $\mu_2^{\Theta}(P, P, S) = 1$, true type perfectly identified.

Factored marginals. Computing each $\mu_2^{\Theta,i}$ independently:

$$\mu_2^{\Theta,1}(P) = 1, \quad \mu_2^{\Theta,2}(P) = 1, \quad \mu_2^{\Theta,3}(S) = 1. \quad (57)$$

The product $\mu_1^{\Theta,1}(P) \cdot \mu_2^{\Theta,2}(P) \cdot \mu_2^{\Theta,3}(S)$ equals $\delta_{(P,P,S)}$, matching the joint posterior exactly. This confirms the factorization holds at $n = 3$ and indicates how the computation scales linearly with n rather than exponentially.

G Verification of $c_k(\tau_k)$ Independence of (P, θ)

We elaborate on the claim in Lemma B.1 that $c_k(\tau_k)$ does not depend on (P, θ) .

The policy π_k is a random variable taking values in $\Delta(\mathcal{A})^S$. Its distribution $P(\pi_k | \mathcal{H}_{k-1})$ is induced by the algorithm:

$$P(\pi_k | \mathcal{H}_{k-1}) = \int \mathbb{1}[\pi_k = \text{CCE}(\hat{P}, \hat{\theta})] d\mu_k^P(\hat{P}) \prod_i d\mu_k^{\Theta,i}(\hat{\theta}_i). \quad (58)$$

Both μ_k^P and $\mu_k^{\Theta,i}$ are measurable with respect to $\mathcal{H}_{k-1}$. By the tower property of conditional expectation, $P(\pi_k | \mathcal{H}_{k-1}, P, \theta) = P(\pi_k | \mathcal{H}_{k-1})$, since the algorithmic randomness in choosing π_k is conditionally independent of the true latent given the history. Hence

$$c_k(\tau_k) = \int P(\pi_k | \mathcal{H}_{k-1}) \prod_{h=1}^H \pi_k(a_h^k | s_h^k) d\pi_k \quad (59)$$

is a function of τ_k and $\mathcal{H}_{k-1}$ alone.

This argument reflects a standard property of posterior sampling: the algorithm uses only the posterior, not the truth, to choose its policy. The truth enters the system only through the realized trajectory.

H Discussion of CCE Oracle Approximation

In Algorithm 1, the CCE oracle is assumed to return an ϵ_{CCE} -approximate CCE. We comment on how this approximation enters the regret bound.

Let π_k be the policy returned by the oracle on the sampled world $(\hat{P}_k, \hat{\theta}_k)$. Then

$$\max_{\pi'} V_i^{\pi'}(s_1; \hat{\theta}_i^k, \hat{P}_k) - V_i^{\pi_k}(s_1; \hat{\theta}_i^k, \hat{P}_k) \leq \epsilon_{\text{CCE}}, \quad \forall i. \quad (60)$$

This per-episode error contributes additively across K episodes, yielding the $K \cdot \epsilon_{\text{CCE}}$ term in (34). Standard V-learning style oracles (Jin et al., 2021) achieve $\epsilon_{\text{CCE}} = O(1/\sqrt{T_{\text{inner}}})$ where T_{inner} is the inner loop budget. Setting $T_{\text{inner}} = \sqrt{K}$ yields $K \cdot \epsilon_{\text{CCE}} = O(\sqrt{K})$, which does not dominate the regret.

I Per-Episode Regret Decomposition: Full Derivation

This appendix provides the rigorous decomposition of per-episode regret into a transition-misspecification component and a reward-misspecification component. The decomposition is the technical bridge between Theorem B.2 (posterior decoupling) and the cumulative regret bound sketched in Section E.

I.1 Notation and Setup

We use the value function notation $V_{i,h}^\pi(s; P, \theta_i)$ for agent i 's expected return from state s at step h under policy π , transition kernel P , and own type θ_i . Formally,

$$V_{i,h}^\pi(s; P, \theta_i) := \mathbb{E}_{\pi, P} \left[\sum_{h'=h}^H r_i(s_{h'}, a_{h'}, \theta_i) \mid s_h = s \right], \quad (61)$$

with $V_{i,H+1}^\pi \equiv 0$. The state-action value satisfies the Bellman recursion

$$Q_{i,h}^\pi(s, a; P, \theta_i) = r_i(s, a, \theta_i) + \mathbb{E}_{s' \sim P(\cdot | s, a)} [V_{i,h+1}^\pi(s'; P, \theta_i)]. \quad (62)$$

The episode- k regret of agent i is

$$\Delta_i^k := V_{i,1}^{\pi^*}(s_1^k; P^*, \theta_i^*) - V_{i,1}^{\pi_k}(s_1^k; P^*, \theta_i^*), \quad (63)$$

where $\pi^* := \text{CCE}(P^*, \theta^*)$ is the equilibrium of the true world and $\pi_k := \text{CCE}(\hat{P}_k, \hat{\theta}_k)$ is the equilibrium of the sampled world.

I.2 Reduction to the Simulation Gap via Posterior Sampling

The first step is to convert the regret expression into a comparison between the sampled and true value of the executed policy. The reduction relies on a property of posterior sampling formalized by Russo and Van Roy (2014); Osband and Van Roy (2017).

Lemma I.1 (Posterior Sampling Equivalence). *Under Assumptions A.5 and A.6, let f be any measurable function on (P, θ, s_1) , possibly depending on $\mathcal{H}_{k-1}$. Then*

$$\mathbb{E}[f(\hat{P}_k, \hat{\theta}_k, s_1^k) \mid \mathcal{H}_{k-1}] = \mathbb{E}[f(P^*, \theta^*, s_1^k) \mid \mathcal{H}_{k-1}]. \quad (64)$$

Proof. By Assumption A.5, $s_1^k \sim \beta$ is independent of (P^*, θ^*) , so conditioning on s_1^k does not affect the posterior over the latent. By construction of H-PSMG (Algorithm 1), $(\hat{P}_k, \hat{\theta}_k)$ are drawn from the posterior $P(P, \theta \mid \mathcal{H}_{k-1})$, which is, by definition, the conditional distribution of (P^*, θ^*) given $\mathcal{H}_{k-1}$. Hence $(\hat{P}_k, \hat{\theta}_k)$ and (P^*, θ^*) have identical conditional distribution. The equality of expectations follows. Assumption A.6 ensures f is well-defined when it depends on $\text{CCE}(P, \theta)$, since the oracle is a deterministic mapping. $\square$

Lemma I.2 (Reduction to Simulation Gap). *The Bayesian regret of agent i at episode k admits the equality*

$$\mathbb{E}[\Delta_i^k \mid \mathcal{H}_{k-1}] = \mathbb{E}[V_{i,1}^{\pi_k}(s_1^k; \hat{P}_k, \hat{\theta}_i^k) - V_{i,1}^{\pi_k}(s_1^k; P^*, \theta_i^*) \mid \mathcal{H}_{k-1}]. \quad (65)$$

Proof. Apply Lemma I.1 with $f(P, \theta) := V_{i,1}^{\text{CCE}(P, \theta)}(s_1^k; P, \theta_i)$:

$$\mathbb{E}[V_{i,1}^{\pi^*}(s_1^k; P^*, \theta_i^*) \mid \mathcal{H}_{k-1}] = \mathbb{E}[V_{i,1}^{\pi_k}(s_1^k; \hat{P}_k, \hat{\theta}_i^k) \mid \mathcal{H}_{k-1}]. \quad (66)$$

Subtracting $\mathbb{E}[V_{i,1}^{\pi_k}(s_1^k; P^*, \theta_i^*) \mid \mathcal{H}_{k-1}]$ from both sides and rewriting the left side using definition (63) yields (65). $\square$

The reduction shifts the problem from comparing the true optimal value to comparing two evaluations of the same policy, one under the sampled world and one under the true world. This is the standard *simulation gap* in posterior sampling analysis.

I.3 Value Difference Lemma for HT-MG

To bound the simulation gap, we adapt the classical simulation lemma (Kearns and Singh, 2002; Bai and Jin, 2020) to HT-MG with the value function indexed by type.

Lemma I.3 (Value Difference for HT-MG). *For any policy π , transition kernels P_1, P_2 , types*

$\theta_{i,1}, \theta_{i,2}$, and initial state s_1 ,

$$V_{i,1}^\pi(s_1; P_1, \theta_{i,1}) - V_{i,1}^\pi(s_1; P_2, \theta_{i,2}) = \sum_{h=1}^H \mathbb{E}_{(s,a) \sim d_h^{\pi, P_1}} [\Phi_h^R(s, a) + \Phi_h^P(s, a)], \quad (67)$$

where the per-step reward and transition discrepancies are

$$\Phi_h^R(s, a) := r_i(s, a, \theta_{i,1}) - r_i(s, a, \theta_{i,2}), \quad (68)$$

$$\Phi_h^P(s, a) := \langle P_1(\cdot | s, a) - P_2(\cdot | s, a), V_{i,h+1}^\pi(\cdot; P_2, \theta_{i,2}) \rangle \quad (69)$$

and d_h^{π, P_1} is the occupancy distribution of (s_h, a_h) under π in P_1 .

Proof. Define the residual $\delta_h(s) := V_{i,h}^\pi(s; P_1, \theta_{i,1}) - V_{i,h}^\pi(s; P_2, \theta_{i,2})$ for $h \in \{1, \dots, H+1\}$. Note $\delta_{H+1} \equiv 0$. For $h \leq H$,

$$\delta_h(s) = \mathbb{E}_{a \sim \pi_h(\cdot | s)} [Q_{i,h}^\pi(s, a; P_1, \theta_{i,1}) - Q_{i,h}^\pi(s, a; P_2, \theta_{i,2})] \quad (70)$$

$$= \mathbb{E}_{a \sim \pi} [r_i(s, a, \theta_{i,1}) - r_i(s, a, \theta_{i,2}) + \mathbb{E}_{s' \sim P_1} [V_{i,h+1}^\pi(s'; P_2, \theta_{i,2}) - V_{i,h+1}^\pi(s'; P_1, \theta_{i,1})]] \quad (71)$$

Add and subtract $\mathbb{E}_{s' \sim P_1} [V_{i,h+1}^\pi(s'; P_2, \theta_{i,2})]$ inside the bracket:

$$\delta_h(s) = \mathbb{E}_{a \sim \pi} [\Phi_h^R(s, a) + \Phi_h^P(s, a) + \mathbb{E}_{s' \sim P_1} [\delta_{h+1}(s')]] \quad (72)$$

Unrolling the recursion from $h = 1$ to H and using $\delta_{H+1} = 0$,

$$\delta_1(s_1) = \sum_{h=1}^H \mathbb{E}_{(s_h, a_h) \sim d_h^{\pi, P_1}} [\Phi_h^R(s_h, a_h) + \Phi_h^P(s_h, a_h)], \quad (73)$$

which is (67). $\square$

Remark I.4 (Symmetric Form with P_2 Rollout). The identity (67) admits a symmetric form obtained by adding and subtracting $\mathbb{E}_{s' \sim P_2} [V_{i,h+1}^\pi(s'; P_1, \theta_{i,1})]$ instead of $\mathbb{E}_{s' \sim P_1} [V_{i,h+1}^\pi(s'; P_2, \theta_{i,2})]$ in the derivation. Unrolling the resulting residual under P_2 yields

$$V_{i,1}^\pi(s_1; P_1, \theta_{i,1}) - V_{i,1}^\pi(s_1; P_2, \theta_{i,2}) = \sum_{h=1}^H \mathbb{E}_{(s,a) \sim d_h^{\pi, P_2}} [\Phi_h^R(s, a) + \tilde{\Phi}_h^P(s, a)] \quad (74)$$

where the rollout occupancy uses P_2 and the transition-discrepancy weight is taken under P_1 :

$$\tilde{\Phi}_h^P(s, a) := \langle P_1(\cdot | s, a) - P_2(\cdot | s, a), V_{i,h+1}^\pi(\cdot; P_1, \theta_{i,1}) \rangle. \quad (75)$$

The symmetric form is the one used to bound the transition-misspec term, as it lets us evaluate the inner product along the realized trajectory drawn from P^* rather than along a hypothetical rollout under $\hat{P}_k$ (see Proposition I.13).

Applying Lemma I.3 with $P_1 = \hat{P}_k$, $P_2 = P^*$, $\theta_{i,1} = \hat{\theta}_i^k$, $\theta_{i,2} = \theta_i^*$, and combining with Lemma I.2 yields the central decomposition:

$$\mathbb{E}[\Delta_i^k | \mathcal{H}_{k-1}] = \underbrace{\mathbb{E} \left[\sum_{h=1}^H \mathbb{E}_{d_h^{\pi_k, \hat{P}_k}} \Phi_h^R \middle| \mathcal{H}_{k-1} \right]}_{\Delta_i^{k,R}} + \underbrace{\mathbb{E} \left[\sum_{h=1}^H \mathbb{E}_{d_h^{\pi_k, \hat{P}_k}} \Phi_h^P \middle| \mathcal{H}_{k-1} \right]}_{\Delta_i^{k,P}} \quad (76)$$

The reward-misspec term $\Delta_i^{k,R}$ is driven by the gap between sampled and true type. The transition-misspec term $\Delta_i^{k,P}$ is driven by the gap between sampled and true transition kernel. Theorem B.2 ensures these two sources of error are governed by independent posteriors.

I.4 Bounding the Reward-Misspecification Term

Since rewards lie in $[0, 1]$, $|\Phi_h^R(s, a)| \leq 1$ everywhere, and $|\Phi_h^R(s, a)| = 0$ whenever $\hat{\theta}_i^k = \theta_i^*$ by Assumption A.2. Therefore

$$\Delta_i^{k,R} \leq \mathbb{E} \left[\sum_{h=1}^H \mathbb{E}_{d_h^{\pi_k, \hat{P}_k}} |\Phi_h^R(s, a)| \middle| \mathcal{H}_{k-1} \right] \leq \mathbb{E} \left[\sum_{h=1}^H \mathbb{E}_{d_h^{\pi_k, \hat{P}_k}} \mathbb{1}[\hat{\theta}_i^k \neq \theta_i^*] \middle| \mathcal{H}_{k-1} \right] \leq \mathbb{E} \left[\sum_{h=1}^H \mathbb{E}_{d_h^{\pi_k, \hat{P}_k}} \|\mu_k^{\Theta, i}\|_2^2 \middle| \mathcal{H}_{k-1} \right] \quad (77)$$

where the last equality uses Lemma I.1 with $f(\theta) = \mathbb{1}[\hat{\theta}_i^k = \theta_i^*]$ and the fact that under posterior sampling, $\hat{\theta}_i^k$ and θ_i^* are conditionally i.i.d. from $\mu_k^{\Theta, i}$. The collision probability $\|\mu_k^{\Theta, i}\|_2^2 = \sum_{\theta} \mu_k^{\Theta, i}(\theta)^2$ is bounded by 1 and approaches 1 as the posterior concentrates.

We now apply the information-theoretic framework of Russo and Van Roy (2016). Define the information ratio at episode k :

$$\Gamma_k^i := \frac{(\mathbb{E}[\Delta_i^{k,R} | \mathcal{H}_{k-1}])^2}{I_k(\theta_i^*; \tau_k)}, \quad (78)$$

where $I_k(\theta_i^*; \tau_k) := H(\mu_k^{\Theta, i}) - \mathbb{E}[H(\mu_{k+1}^{\Theta, i}) | \mathcal{H}_{k-1}]$ is the expected information gain about θ_i^* from observing τ_k , and $H(\cdot)$ denotes Shannon entropy.

Proposition I.5 (Type-Inference Regret via Information Ratio). If $\Gamma_k^i \leq \bar{\Gamma}$ uniformly over $k \leq K$, then

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq \sqrt{\bar{\Gamma} \cdot \log |\Theta_i| \cdot K}. \quad (79)$$

Proof Sketch. By Cauchy-Schwarz applied to $\sum_k \mathbb{E}[\Delta_i^{k,R}] = \sum_k \sqrt{\Gamma_k^i \cdot I_k}$,

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq \sqrt{\sum_k \Gamma_k^i \cdot \sum_k I_k} \leq \sqrt{\bar{\Gamma} K \cdot H_0(\theta_i^*)}, \quad (80)$$

where $\sum_k I_k \leq H_0(\theta_i^*) = H(\mu_1^{\Theta, i}) \leq \log |\Theta_i|$ by the telescoping property of entropy. $\square$

The information ratio $\bar{\Gamma}$ depends on the problem. We bound it for the present setting by reducing the type-inference subproblem to a $|\Theta_i|$ -armed Bayesian

bandit and adapting the information-ratio analysis of [Russo and Van Roy \(2016\)](#).

Lemma I.6 (Information Ratio Bound for HT-MG Type Inference). *Under Assumptions A.1, A.2, A.5, A.6, the per-episode information ratio for type inference satisfies*

$$\Gamma_k^i \leq \frac{H^2 |\Theta_i|}{2}, \quad \forall k \geq 1, \forall i \in [n]. \quad (81)$$

Proof. The proof proceeds in four steps. We first cast type inference as a meta-bandit, then bound per-pair regret by Pinsker’s inequality, apply Cauchy-Schwarz to extract the $|\Theta_i|$ factor, and identify the resulting weighted-KL sum with the conditional mutual information.

Step 1 (Meta-bandit reduction). Fix episode k and condition on $\mathcal{H}_{k-1}$. Write $\nu_k := \mu_k^{\Theta_i, i}$ for the posterior on agent i ’s type. By Theorem B.2 and Algorithm 1, the sampled type $\hat{\theta}_i^k$ and the latent truth θ_i^* are conditionally i.i.d. from ν_k given $\mathcal{H}_{k-1}$. Define the pairwise conditional regret

$$f_k(\hat{\theta}, \theta^*) := \sum_{h=1}^H \mathbb{E}^{\pi_k(\hat{\theta}), P^*} [\mu_i(s_h, a_h, \hat{\theta}) - \mu_i(s_h, a_h, \theta^*)], \quad (82)$$

where $\pi_k(\hat{\theta}) := \text{CCE}(\hat{P}_k, \hat{\theta}, \hat{\theta}_{-i}^k)$ is the executed policy when agent i ’s sampled type is $\hat{\theta}$ (other sampled quantities held fixed), and $\mu_i(s, a, \theta) := \mathbb{E}_{r \sim P(\cdot | s, a, \theta)}[r]$ is the mean reward of type θ . By Assumption A.2, $f_k(\theta, \theta) = 0$, and the bound $r \in [0, 1]$ gives $|f_k(\hat{\theta}, \theta^*)| \leq H$ uniformly. Then

$$\mathbb{E}[\Delta_i^{k,R} | \mathcal{H}_{k-1}] = \mathbb{E}_{\hat{\theta}, \theta^* \sim \nu_k} [f_k(\hat{\theta}, \theta^*)] = \sum_{\theta^* \in \Theta_i} \nu_k(\theta^*) \mathbb{E}_{\hat{\theta} \sim \nu_k} [f_k(\hat{\theta}, \theta^*)] \quad (83)$$

This is the Bayesian-bandit-style decomposition with arms Θ_i , sampled action $\hat{\theta}_i^k$, optimal action θ_i^* , and pairwise regret f_k .

Step 2 (Per-pair Pinsker bound). For each $(s, a, \hat{\theta}, \theta^*)$, since the reward $r \in [0, 1]$,

$$|\mu_i(s, a, \hat{\theta}) - \mu_i(s, a, \theta^*)| \leq \|P_i(\cdot | s, a, \hat{\theta}) - P_i(\cdot | s, a, \theta^*)\|_1 \quad (84)$$

where the first inequality bounds a difference of means by total variation between reward distributions, and the second is Pinsker’s inequality. Squaring (82) and applying Cauchy-Schwarz over the H terms in the sum,

$$f_k(\hat{\theta}, \theta^*)^2 \leq H \sum_{h=1}^H \mathbb{E}^{\pi_k(\hat{\theta}), P^*} [(\mu_i(s_h, a_h, \hat{\theta}) - \mu_i(s_h, a_h, \theta^*))^2] \quad (85)$$

$$\leq \frac{H}{2} \sum_{h=1}^H \mathbb{E}^{\pi_k(\hat{\theta}), P^*} [D_{\text{KL}}(P_i(\cdot | s_h, a_h, \hat{\theta}) \| P_i(\cdot | s_h, a_h, \theta^*))] \quad (86)$$

$$= \frac{H}{2} \cdot D_{\text{KL}}(P_{\hat{\theta}}^\tau \| P_{\theta^*}^\tau), \quad (87)$$

where the last equality applies the chain rule of relative entropy: $P_{\hat{\theta}}^\tau$ denotes the trajectory distribution including agent i ’s reward observations under reward type θ , with policy $\pi_k(\hat{\theta})$ and other randomness fixed; under Assumption A.2 only the agent- i reward factor depends on θ , so $D_{\text{KL}}(P_{\hat{\theta}}^\tau \| P_{\theta^*}^\tau)$ telescopes into the per-step KL terms.

Step 3 (Cauchy-Schwarz over θ^ with $|\Theta_i|$ factor).* From (83),

$$(\mathbb{E}[\Delta_i^{k,R} | \mathcal{H}_{k-1}])^2 = \left(\sum_{\theta^* \in \Theta_i} \nu_k(\theta^*) \cdot g_k(\theta^*) \right)^2. \quad (88)$$

Apply the standard Cauchy-Schwarz inequality on the inner product $\sum_{\theta^*} 1 \cdot (\nu_k(\theta^*) g_k(\theta^*))$:

$$\left(\sum_{\theta^*} \nu_k(\theta^*) g_k(\theta^*) \right)^2 \leq |\Theta_i| \cdot \sum_{\theta^*} \nu_k(\theta^*)^2 g_k(\theta^*)^2 \leq |\Theta_i| \cdot \sum_{\theta^*} \nu_k(\theta^*) g_k(\theta^*) \quad (89)$$

using $\nu_k(\theta^*) \in [0, 1]$ in the second inequality. By Jensen’s inequality applied to $g_k(\theta^*) = \mathbb{E}_{\hat{\theta}}[f_k(\hat{\theta}, \theta^*)]$,

$$(\hat{\theta}, \theta^*) \in \Theta_i^2, \quad \mathbb{E}_{\hat{\theta} \sim \nu_k} [f_k(\hat{\theta}, \theta^*)^2] \leq \mathbb{E}_{\hat{\theta} \sim \nu_k} [f_k(\hat{\theta}, \theta^*)]^2. \quad (90)$$

Combining with the Pinsker bound (87),

$$(\mathbb{E}[\Delta_i^{k,R} | \mathcal{H}_{k-1}])^2 \leq \frac{H |\Theta_i|}{2} \cdot \mathbb{E}_{\hat{\theta}, \theta^* \sim \nu_k} [D_{\text{KL}}(P_{\hat{\theta}}^\tau \| P_{\theta^*}^\tau)]. \quad (91)$$

Step 4 (Identification with conditional mutual information). It remains to show

$$\mathbb{E}_{\hat{\theta}, \theta^*} [D_{\text{KL}}(P_{\hat{\theta}}^\tau \| P_{\theta^*}^\tau)] \leq H \cdot I_k(\theta_i^*; \tau_k | \mathcal{H}_{k-1}), \quad (92)$$

where I_k is the conditional mutual information between θ_i^* and the trajectory τ_k given $\mathcal{H}_{k-1}$.

By Assumption A.2 the trajectory distribution $P_{\hat{\theta}}^\tau$ factorizes as $P_{\hat{\theta}}^\tau(\tau) = P_{\text{base}}^\tau(s_{1:H+1}, a_{1:H}) \cdot \prod_h P_i(r_h^i | s_h, a_h, \theta) \cdot Q^{-i}(r_{1:H}^{-i} | \cdot)$ where P_{base}^τ and Q^{-i} are independent of agent i ’s reward type. Therefore $D_{\text{KL}}(P_{\hat{\theta}}^\tau \| P_{\theta^*}^\tau)$ reduces to the agent- i reward channel. By the chain rule of relative entropy and standard mutual information identities

([Russo and Van Roy, 2006](#), Section 5.3), also [Lattimore and Szepesvári, 2020](#), Theorem 36.5),

$\mathbb{E}_{\hat{\theta}}[D_{\text{KL}}(P_{\hat{\theta}}^\tau \| \bar{P}^\tau)] = I_k$ where $\bar{P}^\tau = \mathbb{E}_{\theta \sim \nu_k}[P_{\theta}^\tau]$. Combined with the convexity of D_{KL} in its second argument ([Cover and Thomas, 2006](#), Theorem 2.7.2), $\mathbb{E}_{\theta^*}[D_{\text{KL}}(P_{\hat{\theta}}^\tau \| P_{\theta^*}^\tau)] \leq C(\nu_k)$ for a problem-dependent constant. Adapting Lemma 1 and Proposition 5 of [Russo and Van Roy \(2016\)](#), which carry through verbatim when the bandit reward is replaced by trajectory observation under their general feedback model, yields (92) with the factor H accounting for trajectory horizon as in their Section

Substituting (92) into (91),

$$(\mathbb{E}[\Delta_i^{k,R} | \mathcal{H}_{k-1}])^2 \leq \frac{H^2 |\Theta_i|}{2} \cdot I_k(\theta_i^*; \tau_k | \mathcal{H}_{k-1}), \quad (93)$$

from which $\Gamma_k^i \leq H^2 |\Theta_i|/2$. $\square$

Remark I.7 (Tightness of the $|\Theta_i|$ factor). *The $|\Theta_i|$ factor in the information-ratio bound is the standard worst-case rate for finite-armed Thompson sampling (Russo and Van Roy, 2016). It can be improved under additional structure: under Assumption A.4, posterior concentration is geometric (Lemma I.10), and the cumulative regret reduces from $\sqrt{|\Theta_i|K \log |\Theta_i|}$ to $|\Theta_i|/\rho$ (Corollary I.11). For continuous type spaces, the $|\Theta_i|$ factor must be replaced by the relevant covering or eluder dimension; see Russo and Van Roy (2014) for the eluder-dimension version of the bound.*

Remark I.8 (Why trajectory feedback is at least as informative as scalar feedback). *In standard K -armed bandit Thompson sampling, the feedback Y_t is the realized scalar reward $r_t \in [0, 1]$. For our setting, $Y_k = \tau_k$ is the full trajectory, which by data-processing inequality is at least as informative about θ_i^* as the agent’s marginal reward history. Hence $I_k(\theta_i^*; \tau_k) \geq I_k(\theta_i^*; r_k)$, the information ratio under trajectory feedback is at most that under scalar feedback, and the bound $\Gamma_k^i \leq H^2 |\Theta_i|/2$ is preserved. In practice trajectory feedback yields a tighter information ratio whenever the trajectory carries information about θ_i^* beyond marginal rewards (e.g., through reward variance or state visitation patterns).*

Remark I.9 (Multi-agent CCE adaptation). *The information-ratio framework of Russo and Van Roy (2016) was originally developed for single-agent posterior sampling in bandit and MDP settings. The proof of Lemma I.6 adapts it to per-agent regret in the CCE-oracle setting via the following pipeline. (i) Theorem B.2 (decoupling) reduces the joint posterior to the product of per-agent marginals, so $\hat{\theta}_i^k$ and θ_i^* are conditionally i.i.d. from the same marginal $\nu_k = \mu_k^{\Theta_i}$ given $\mathcal{H}_{k-1}$. (ii) The pairwise regret $f_k(\hat{\theta}, \theta^*)$ in Step 1 treats $\pi_k(\hat{\theta}) = \text{CCE}(\hat{P}_k, \hat{\theta}, \hat{\theta}_{-i}^k)$ as a deterministic mapping (Assumption A.6), so the CCE oracle’s joint-policy nature is absorbed into a single $\hat{\theta}_i$ -dependent policy. (iii) Assumption A.2 (own-type reward locality) makes the trajectory KL telescope into per-step per-agent KL on agent i ’s reward likelihood, eliminating cross-agent coupling in the information accounting. The three ingredients together convert the multi-agent CCE problem into a per-agent meta-bandit on Θ_i , on which the standard Russo-Van Roy bound applies verbatim. Xiong et al. (2022) develop a related multi-agent decoupling coefficient for zero-sum function-approximation MGs; our analysis differs in that the decoupling comes from the structural assumptions TI + RL + PF rather than from a function-class complexity measure.*

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq O(H \sqrt{|\Theta_i| \log |\Theta_i| \cdot K}). \quad (94)$$

This is the minimax (gap-independent) rate. Under TID a sharper rate is available.

Lemma I.10 (Posterior Concentration under TID). *Under Assumptions A.4 and A.5, the posterior collision probability decays geometrically:*

$$1 - \|\mu_k^{\Theta_i}\|_2^2 \leq |\Theta_i| \cdot \exp(-\rho \cdot N_k^{\text{eff}}), \quad (95)$$

where $N_k^{\text{eff}} := \sum_{j < k} \mathbb{E}[N_j(s, a) \mid (s, a) \in \text{supp}(d^{\pi_j, P^*})]$ is the effective number of informative state-action visits up to episode k . In particular, $N_k^{\text{eff}} = \Omega(k)$ under any policy that visits informative states with positive probability.

Proof Sketch. By Assumption A.4, every informative (s, a) visit produces a Hellinger-separated reward distribution between any two candidate types. Standard Bayesian concentration arguments (Ghosal and van der Vaart, 2017, Theorem 4.2) then give exponential decay of posterior mass on non-true types at rate ρ per informative observation. Summing the geometric series and taking a union bound over $|\Theta_i| - 1$ alternative types yields the stated bound. $\square$

Corollary I.11 (Gap-Dependent Type Regret). *Under Assumptions A.4, A.5, A.6, the cumulative type-misspec regret is bounded independently of K :*

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] \leq H \sum_{k=1}^K (1 - \|\mu_k^{\Theta_i}\|_2^2) \leq \frac{H \cdot |\Theta_i|}{1 - \exp(-\rho)} \leq O\left(\frac{H |\Theta_i|}{\rho}\right). \quad (96)$$

For small ρ , $1 - \exp(-\rho) \approx \rho$, recovering the form $O(H |\Theta_i|/\rho)$. Replacing $|\Theta_i|$ by $\log |\Theta_i|$ via the union-bound step in Lemma I.10 yields $O(H \log |\Theta_i|/\rho)$, the form stated in Corollary E.2 of the main text.

Proof. The first inequality follows from Eq. (31) of the main text: $|\Delta_i^{k,R}| \leq H \cdot \mathbb{1}[\hat{\theta}_i^k \neq \theta_i^*]$ in conditional expectation. The second applies Lemma I.10 and sums the geometric series $\sum_{k=1}^{\infty} \exp(-\rho \cdot N_k^{\text{eff}}) \leq \sum_{k=1}^{\infty} \exp(-\rho k) = 1/(1 - \exp(-\rho))$. $\square$

Remark I.12 (Interpretation). *The constant-in- K bound under TID is consistent with the intuition that finite-type inference is a one-shot problem: once the posterior identifies the true type with high confidence, no further type-misspec error accumulates. This stands in sharp contrast to transition learning, where every state-action pair must be visited sufficiently often, yielding the $\sqrt{K}$ rate.*

I.5 Bounding the Transition-Misspecification Term

The transition-misspec term is bounded by combining Hölder’s inequality with Dirichlet posterior concentration. Throughout we assume the standard PSRL prior μ_1^P is a product of $|\mathcal{S}| \cdot |\mathcal{A}|$ symmetric Dirichlet distributions, $\mu_1^P(\cdot | s, a) = \text{Dir}(\alpha_0 \mathbf{1})$ for some concentration parameter $\alpha_0 > 0$.

For convenience, write $V_{h+1} := V_{i,h+1}^{\pi_k}(\cdot; P^*, \hat{\theta}_i^k)$, suppressing the (i, k, π_k) dependence. Note $\|V_{h+1}\|_\infty \leq H$ since rewards lie in $[0, 1]$ and the horizon is H .

We present two complementary bounds. Proposition I.13 gives a fully rigorous Hoeffding-level derivation with explicit constants. Proposition I.14 states the sharper Bernstein-refined bound, citing the standard variance-aware analysis of [Lu and Van Roy \(2019\)](#); [Azar et al. \(2017\)](#).

Proposition I.13 (Transition-Learning Regret, Hoeffding-Level Bound). *Under the Dirichlet prior on P^* and posterior sampling, the cumulative expected transition-misspec regret satisfies*

$$\sum_{k=1}^K \mathbb{E}[\|\Delta_i^{k,P}\|] \leq C_1 \cdot H^{3/2} \cdot S \cdot \sqrt{A_{\text{joint}} K \log(SA_{\text{joint}}KH)}, \quad (97)$$

where $A_{\text{joint}} := \prod_{j=1}^n |\mathcal{A}_j|$ and $C_1 > 0$ is a universal constant.

Proof. We carry out the proof in five steps.

Step 0 (Switch to P^ rollout via symmetric simulation lemma).* The central decomposition (76) writes $\Delta_i^{k,P}$ as an expectation under $d_h^{\pi_k, \hat{P}_k}$, i.e., rollout under the sampled kernel. This occupancy is not the realized trajectory and does not align with the visit counts $N_k(s, a)$ that drive the pigeonhole bound. By Remark I.4 (the symmetric form of Lemma I.3 with $P_1 = \hat{P}_k$, $P_2 = P^*$), the same value gap admits the equivalent representation

$$\Delta_i^{k,P} = \mathbb{E}^{\pi_k, P^*} \left[\sum_{h=1}^H \langle (\hat{P}_k - P^*)(\cdot | s_h, a_h), \tilde{V}_{h+1} \rangle \right], \quad (98)$$

where the rollout occupancy now matches the realized trajectory $(s_h^k, a_h^k)_{h=1}^H \sim d^{\pi_k, P^*}$ at episode k . The value-to-go $\tilde{V}_{h+1}$ is evaluated under the sampled MDP, and $\|\tilde{V}_{h+1}\|_\infty \leq H$ since rewards lie in $[0, 1]$.

Step 1 (Per-step Hölder bound). By Hölder’s inequality applied to the inner product in (98) and the value bound $\|\tilde{V}_{h+1}\|_\infty \leq H$,

$$|\langle (\hat{P}_k - P^*)(\cdot | s, a), \tilde{V}_{h+1} \rangle| \leq H \cdot \|\hat{P}_k(\cdot | s, a) - P^*(\cdot | s, a)\|_1. \quad (99)$$

Step 2 (ℓ_1 concentration of Dirichlet posterior sample to truth). Let $N_k(s, a) := |\{(t, h) : t < k, h \in [H], s_h^t = s, a_h^t = a\}|$ be the visit count

to (s, a) accumulated across all prior episodes and let $\bar{P}_k(\cdot | s, a) := \mathbb{E}[\hat{P}_k(\cdot | s, a) | \mathcal{H}_{k-1}]$ denote the posterior mean. Decompose via triangle inequality:

$$\|\hat{P}_k(\cdot | s, a) - P^*(\cdot | s, a)\|_1 \leq \|\hat{P}_k(\cdot | s, a) - \bar{P}_k(\cdot | s, a)\|_1 + \|\bar{P}_k(\cdot | s, a) - P^*(\cdot | s, a)\|_1 \quad (100)$$

The posterior mean satisfies $\bar{P}_k(\cdot | s, a) = (\alpha_0 \mathbf{1} + N_k(s, a, \cdot)) / (S\alpha_0 + N_k(s, a))$, where $N_k(s, a, s')$ is the count of transitions (s, a, s') in $\mathcal{H}_{k-1}$. Writing $\hat{P}_k^{\text{emp}}(\cdot | s, a) := N_k(s, a, \cdot) / N_k(s, a)$ for the empirical mean, the bias from the prior pseudo-counts is bounded by

$$\|\bar{P}_k(\cdot | s, a) - \hat{P}_k^{\text{emp}}(\cdot | s, a)\|_1 \leq \frac{S\alpha_0}{N_k(s, a) + S\alpha_0} = O\left(\frac{S\alpha_0}{N_k(s, a) + 1}\right) \quad (101)$$

which is dominated by the $O(\sqrt{S/N})$ term derived below. We absorb it into the universal constant.

For the empirical mean to truth: by [Weissman et al. \(2003\)](#), for any $\delta_1 \in (0, 1)$,

$$P\left(\|\hat{P}_k^{\text{emp}}(\cdot | s, a) - P^*(\cdot | s, a)\|_1 > \sqrt{\frac{2S \log(2/\delta_1)}{N_k(s, a)}} \mid \mathcal{H}_{k-1}, N_k(s, a)\right) \leq \delta_1 \quad (102)$$

For the posterior sample to its mean: the Dirichlet posterior $\hat{P}_k(\cdot | s, a) \sim \text{Dir}(\alpha_0 \mathbf{1} + N_k(s, a, \cdot))$ concentrates around its mean at the same ℓ_1 rate. By standard moment computation for Dirichlet linear functionals ([Lattimore and Szepesvári, 2020](#), Section 36.2), $\text{Var}_{\hat{P}_k \sim \text{Dir}}[\hat{P}_k(s' | s, a)] \leq \bar{P}_k(s' | s, a) / (N_k(s, a) + S\alpha_0 + 1)$. Combined with Bennett-type tail bounds for sub-exponential random vectors (see, e.g., [Russo and Van Roy, 2014](#), Section 4),

$$P\left(\|\hat{P}_k(\cdot | s, a) - \bar{P}_k(\cdot | s, a)\|_1 > \sqrt{\frac{2S \log(2/\delta_2)}{N_k(s, a) + 1}} \mid \mathcal{H}_{k-1}\right) \leq \delta_2. \quad (103)$$

Setting $\delta_1 = \delta_2 = (KHS A_{\text{joint}})^{-2}$ and taking a union bound over $(k, s, a) \in [K] \times \mathcal{S} \times \mathcal{A}$, combining (100)-(103) yields: with probability at least $1 - 2(KHS A_{\text{joint}})^{-1}$,

$$\|\hat{P}_k(\cdot | s, a) - P^*(\cdot | s, a)\|_1 \leq C_0 \sqrt{\frac{S \log(KHS A_{\text{joint}})}{N_k(s, a) + 1}} =: \beta_k(s, a), \quad (104)$$

for a universal constant $C_0 > 0$.

Step 3 (Pigeonhole bound on the realized trajectory). On the high-probability event from Step 2, combining (99) with (104) and (98) (which places the expectation under the realized trajectory),

$$\sum_{k=1}^K |\Delta_i^{k,P}| \leq H \sum_{k=1}^K \sum_{h=1}^H \beta_k(s_h^k, a_h^k). \quad (105)$$

By the standard pigeonhole bound on visit counts in tabular RL ([Jaksch et al., 2010](#), Lemma 19),

$$\sum_{k=1}^K \sum_{h=1}^H \frac{1}{\sqrt{N_k(s_h^k, a_h^k) + 1}} \leq (\sqrt{2} + 1) \sqrt{SA_{\text{joint}}HK}. \quad (106)$$

Step 4 (Combining on the high-probability event). On the event of (104),

$$\begin{aligned} \sum_{k=1}^K |\Delta_i^{k,P}| &\leq H \cdot C_0 \sqrt{S \log(KHSA_{\text{joint}})} \cdot \sum_{k,h} \frac{1}{\sqrt{N_k(s_h^k, a_h^k) + 1}} \\ &\leq H \cdot C_0 \sqrt{S \log(KHSA_{\text{joint}})} \cdot (\sqrt{2} + 1) \sqrt{SA_{\text{joint}}HK} \\ &\leq C'_1 \cdot H^{3/2} \cdot S \cdot \sqrt{A_{\text{joint}}K \log(KHSA_{\text{joint}})}. \end{aligned} \quad (107) \quad (108) \quad (109)$$

Step 5 (Bounding the failure event). On the complementary low-probability event of mass $\leq 2/(KHSA_{\text{joint}})$, the per-episode transition-misspec regret is trivially bounded by $|\Delta_i^{k,P}| \leq 2H$ (using $\|\hat{P}_k - P^*\|_1 \leq 2$ in (99)). Summing across K episodes, the failure-event contribution is bounded by

$$2H \cdot K \cdot \frac{2}{KHSA_{\text{joint}}} = \frac{4}{SA_{\text{joint}}}, \quad (110)$$

which is negligible compared to the main term. Taking expectation and absorbing the constant $4/(SA_{\text{joint}})$ into C_1 ,

$$\sum_{k=1}^K \mathbb{E}|\Delta_i^{k,P}| \leq C_1 \cdot H^{3/2} S \sqrt{A_{\text{joint}}K \log(SA_{\text{joint}}KH)}, \quad (111)$$

which is (97). $\square$

The Hoeffding-level proof above contains an extra factor of $\sqrt{S}$ compared to the Bernstein-refined bound. The improvement to $\sqrt{S}$ requires replacing the crude ℓ_1 -vs- ℓ_∞ Hölder argument in Step 1 with a variance-aware bound, then exploiting the law of total variance to control the cumulative variance term. We state the result, citing the established technique.

Proposition I.14 (Transition-Learning Regret, Bernstein-Refined Bound). *Under the same conditions as Proposition I.13, the cumulative expected transition-misspec regret further improves to*

$$\sum_{k=1}^K \mathbb{E}[\|\Delta_i^{k,P}\|] \leq C_2 \cdot H^{3/2} \sqrt{S \cdot A_{\text{joint}} \cdot K \cdot \log(SA_{\text{joint}}KH)} \cdot \mathbb{E}_{\pi_k, P^*} \left[\sum_{h=1}^H \text{Var}_{P^*}(\tilde{V}_{h+1})(s_h, a_h) \right] = \text{Var}^{\pi_k, P^*}(\tilde{V}_1(s_1^k)) \leq \|\tilde{V}_1\|_\infty^2 \leq \dots \quad (112)$$

for a universal constant $C_2 > 0$.

Proof Sketch. The argument follows the variance-aware Bayesian Bernstein analysis of Lu and Van Roy (2019), adapted to the HT-MG setting. We outline the four modifications relative to Proposition I.13.

(0) Use the symmetric simulation lemma. As in Step 0 of Proposition I.13, switch to (98) so

that the expectation is under the realized trajectory $(s_h^k, a_h^k) \sim d^{\pi_k, P^*}$.

(i) Replace Step 1 by the Bernstein-type inner-product bound. For the posterior sample $\tilde{P}_k$ vs. the true kernel P^* , decompose

$$|\langle (\hat{P}_k - P^*)(\cdot | s, a), \tilde{V}_{h+1} \rangle| \leq \underbrace{|\langle (\hat{P}_k - \bar{P}_k)(\cdot | s, a), \tilde{V}_{h+1} \rangle|}_{\text{posterior sample to mean}} + \underbrace{|\langle (\bar{P}_k - P^*)(\cdot | s, a), \tilde{V}_{h+1} \rangle|}_{\text{posterior}} \quad (113)$$

where $\bar{P}_k = \mathbb{E}[\hat{P}_k | \mathcal{H}_{k-1}]$ is the posterior mean.

For the first term, the Dirichlet posterior linear functional $\langle \hat{P}_k, \tilde{V}_{h+1} \rangle$ has variance

$$\text{Var}_{\hat{P}_k \sim \text{Dir}}[\langle \hat{P}_k, \tilde{V}_{h+1} \rangle | \mathcal{H}_{k-1}] = \frac{\text{Var}_{\bar{P}_k}(\tilde{V}_{h+1})}{N_k(s, a) + S\alpha_0 + 1} \quad (114)$$

by the formula for Dirichlet linear functional variance. By Bennett-type sub-exponential tail bounds (Lu and Van Roy, 2019, Theorem 3),

$$|\langle (\hat{P}_k - \bar{P}_k)(\cdot | s, a), \tilde{V}_{h+1} \rangle| \leq c \sqrt{\frac{\text{Var}_{\bar{P}_k}(\tilde{V}_{h+1}) \cdot \log(\text{poly}/\delta)}{N_k(s, a) + 1}} + c \frac{H \log(\dots)}{N_k(s, a)} \quad (115)$$

with probability $1 - \delta$.

For the second term, the posterior mean $\bar{P}_k$ is close to the empirical mean $\hat{P}_k^{\text{emp}}$ (prior bias $O(S\alpha_0/N)$, absorbed into constants). Applying empirical Bernstein (Azar et al., 2017, Lemma 3) to $\hat{P}_k^{\text{emp}}$ vs. P^* ,

$$|\langle (\bar{P}_k - P^*)(\cdot | s, a), \tilde{V}_{h+1} \rangle| \leq c \sqrt{\frac{\text{Var}_{P^*}(\tilde{V}_{h+1}) \cdot \log(\text{poly}/\delta)}{N_k(s, a) + 1}} + c \frac{H \log(\dots)}{N_k(s, a)} \quad (116)$$

with probability $1 - \delta$. Since $|\text{Var}_{\bar{P}_k}(\tilde{V}_{h+1}) - \text{Var}_{P^*}(\tilde{V}_{h+1})| \leq H^2 \|\bar{P}_k - P^*\|_1 = O(H^2 \sqrt{S/N})$ is lower order, (115) and (116) yield, on a high-probability event of mass $1 - 2/(KHSA_{\text{joint}})$,

$$|\langle (\hat{P}_k - P^*)(\cdot | s, a), \tilde{V}_{h+1} \rangle| \leq c' \sqrt{\frac{\text{Var}_{P^*}(\tilde{V}_{h+1}) \cdot \log(SA_{\text{joint}}KH)}{N_k(s, a) + 1}} + c' \dots \quad (117)$$

(ii) Apply the law of total variance to the trajectory under P^* . The realized trajectory $(s_h^k, a_h^k)_{h=1}^H$ is generated by π_k in the true MDP P^* . By the law of total variance,

$$\mathbb{E}_{\pi_k, P^*} \left[\sum_{h=1}^H \text{Var}_{P^*}(\tilde{V}_{h+1})(s_h, a_h) \right] = \text{Var}^{\pi_k, P^*}(\tilde{V}_1(s_1^k)) \leq \|\tilde{V}_1\|_\infty^2 \leq \dots \quad (118)$$

where we used $\|\tilde{V}_{h+1}\|_\infty \leq H$. The crucial point is that the value-to-go $\tilde{V}_{h+1}$ used in the inner product is evaluated under $\tilde{P}_k$ (from (98)), while the variance and rollout are under P^* ; the LoTV identity applies because $\tilde{V}_{h+1}$ is a fixed function of state once $\hat{P}_k, \hat{\theta}_i^k$ are sampled.

(iii) Apply Cauchy-Schwarz across (k, h) on the realized trajectory. The dominant term in

(117) contributes

Azar et al. (2017),

$$\sum_{k=1}^K \sum_{h=1}^H \sqrt{\frac{\text{Var}_{P^*}(\tilde{V}_{h+1})(s_h^k, a_h^k)}{N_k(s_h^k, a_h^k) + 1}} \leq \sqrt{\sum_{k,h} \text{Var}_{P^*}(\tilde{V}_{h+1})(s_h^k, a_h^k)} \sqrt{\sum_{k,h} \frac{1}{N_k(s_h^k, a_h^k) + 1}} \leq \tilde{O}\left(H^{3/2} \sqrt{S \cdot A_{\text{joint}} \cdot K}\right) + O\left(H \sqrt{|\Theta_i| \log |\Theta_i| \cdot K}\right) \quad (119)$$

$$\leq \sqrt{KH^2} \cdot \sqrt{SA_{\text{joint}} \log(KH)} \stackrel{\text{Take expectations in (76)}}{=} H \sqrt{SA_{\text{joint}} K \log(KH)}, \text{ and sum from } k=1 \text{ to } K: \quad (120)$$

where the first inequality is Cauchy-Schwarz, the variance sum is bounded by KH^2 via LoTV in step (ii) applied to each episode, and the harmonic-sum bound $\sum_{k,h} 1/(N_k + 1) \leq SA_{\text{joint}} \log(KH) + SA_{\text{joint}}$ is the standard pigeonhole on $1/N$ (rather than $1/\sqrt{N}$); see, e.g., Lattimore and Szepesvári (2020, Section 3.3).

The lower-order term in (117) contributes

$$\sum_{k,h} \frac{H \log}{N_k + 1} \leq H \log \cdot SA_{\text{joint}} \log(KH) = \tilde{O}(H \cdot SA_{\text{joint}}), \quad (121)$$

which is dominated for $K \gtrsim S$.

Combining. Substituting into (98) and absorbing the failure event as in Step 5 of Proposition I.13,

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^{k,P}] \leq C_2 \cdot H \cdot H \sqrt{SA_{\text{joint}} K \log(SA_{\text{joint}} KH)} + \tilde{O}(H \cdot SA_{\text{joint}}) \quad (122)$$

which is (112). The extension from single-agent PSRL to HT-MG is mechanical: under Assumption A.1, the transition learning problem for the joint kernel $P^*(\cdot | s, a)$ is identical to single-agent PSRL on the product MDP with state space $\mathcal{S}$ and joint action space $\mathcal{A} = \prod_i \mathcal{A}_i$. Theorem B.2 guarantees this reduction is exact, not merely an upper-bound abstraction. $\square$

I.6 Main Regret Theorem and Combination

Combining Propositions I.5, I.13, and I.14 with the central decomposition (76) and accounting for the CCE oracle approximation, we obtain the main result.

Theorem I.15 (Main Regret Bound for H-PSMG). *Assume Assumptions A.1, A.2, A.3, A.5, A.6, that the transition prior is Dirichlet, and that the type information ratio satisfies $\Gamma_k^i \leq H^2 |\Theta_i|/2$ for all $k \in [K]$ and $i \in [n]$. If the CCE oracle returns ϵ_{CCE} -approximate equilibria, then the Bayesian regret of agent i over K episodes admits two bounds:*

Hoeffding-level bound. *Using only Hölder's inequality and Dirichlet ℓ_1 concentration,*

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}\left(H^{3/2} \cdot S \cdot \sqrt{A_{\text{joint}} K}\right) + O\left(H \sqrt{|\Theta_i| \log |\Theta_i| \cdot K}\right) \quad (123)$$

Bernstein-refined bound. *Applying the variance-aware analysis of Lu and Van Roy (2019);*

$$\sum_{k=1}^K \mathbb{E}[\Delta_i^k] = \sum_{k=1}^K \mathbb{E}[\Delta_i^{k,R}] + \sum_{k=1}^K \mathbb{E}[\Delta_i^{k,P}] + \mathcal{E}_{\text{CCE}}, \quad (125)$$

where $\mathcal{E}_{\text{CCE}} \leq K \epsilon_{\text{CCE}}$ accumulates the per-episode equilibrium approximation error (see Appendix H). Applying Proposition I.5 to the type-misspec term and Proposition I.13 (resp. Proposition I.14) to the transition-misspec term yields (123) (resp. (124)). $\square$

I.7 Interpretation of the Bound

The bound (124) has three components, each tied to a distinct source of uncertainty.

The transition-learning component $\tilde{O}(H^{3/2} \sqrt{S \cdot A_{\text{joint}} \cdot K})$ matches single-agent PSRL applied to the joint MDP (Osband et al., 2013). The dependence on $\sqrt{SA_{\text{joint}} K}$ is inherent to MG learning and cannot be improved without additional structural assumptions on the dynamics (e.g., factored transitions).

The type-inference component $O(H \sqrt{|\Theta_i| \log |\Theta_i| \cdot K})$ is independent of the state size S . It captures the difficulty of identifying agent i 's relevant unknown, which is only $\theta_i \in \Theta_i$ and not the full type profile. This is a direct consequence of the agent-level decoupling established in Theorem B.2.

The CCE oracle term $K \cdot \epsilon_{\text{CCE}}$ does not dominate provided $\epsilon_{\text{CCE}} = O(K^{-1/2})$, which is achievable by standard equilibrium solvers (Jin et al., 2021).

Remark I.16 (Improved Rate under Information-Rate Assumption). *If the type information rate is bounded away from zero, i.e., $I_k(\theta_i^*; \tau_k) \geq \rho > 0$ whenever $\hat{\theta}_i^k \neq \theta_i^*$, the type term collapses to $O(H \log |\Theta_i|/\rho)$, constant in K . In this regime the overall bound is dominated by transition learning, and H-PSMG matches the rate of single-agent PSRL on the joint action space.*

Remark I.17 (Comparison to PSRL-ZSG). *For two-player zero-sum MGs, Jafarnia-Jahromi et al. (2021) obtain $\tilde{O}(HS\sqrt{AT})$ Bayesian regret. Specializing Theorem I.15 to $n = 2$, $|\Theta_i| = 1$ (no type uncertainty), and substituting $T = HK$ gives a bound of the same order in K , demonstrating that the type-inference component is a strict additional cost incurred only under type uncertainty.*

J Formal Proof of Theorem E.4: Exponential-in- $|\Theta|^n$ Burn-in Separation

This appendix gives the formal proof of Theorem E.4. We work out the backward-induction algebra on the deep-trap family $\mathcal{G}_m^{\text{trap}}$, establish a matching lower bound for vanilla H-PSMG, and verify the exponential-in- $|\Theta|^n$ separation.

J.1 Setup and Notation

Recall the deep-trap family $\mathcal{G}_m^{\text{trap}}$:

- n agents, $|\Theta_i| = m$ types per agent, $S = m + 2$ states $\{0, 1, \dots, S-1\}$, two actions $A = \{L, R\}$ per agent.
- Joint actions are encoded as integers $0, 1, \dots, 2^n - 1$, with $a^L := 0 = (L, \dots, L)$ and $a^R := 2^n - 1 = (R, \dots, R)$.
- Transitions are deterministic unanimity-vote: $s \rightarrow s+1$ if all agents play R (capped at $S-1$); $s \rightarrow s-1$ if all play L (capped at 0); $s \rightarrow s$ for any mixed joint action.
- Rewards: $r_i(s^*, a, \theta_{\text{jack}}) = 1$ for the rightmost state $s^* = S-1$ and the distinguished jackpot type θ_{jack} ; $r_i(s, a, \theta) = 0.5$ otherwise.
- Initial state $s_1 = 0$; true types $\theta_i^* = \theta_{\text{jack}}$ for all i ; uniform prior $\mu_0^{\Theta, i} = \text{Unif}(\Theta_i)$.
- Horizon H satisfying $H \geq S$, so the oracle policy reaches s^* and accumulates at least one jackpot reward.

We write $p_k := \mu_k^{\Theta, i}(\theta_{\text{jack}})$ for the marginal jackpot mass at episode k (identical across agents by symmetry). At episode k before any informative observation has occurred, $p_k = 1/m$.

For the analysis we adopt the following CCE oracle convention, consistent with the implementation:

Assumption J.1 (CCE Oracle Tie-Breaking). *At each (s, h) , the CCE oracle returns the welfare-maximising CCE among $\{a^L, a^R\}$ if one strictly dominates; if both equal in social welfare $\sum_i V_h^i(s, a)$, the oracle returns a^L (the lowest-index joint action).*

This is the natural deterministic tie-break implemented by $\arg \max$ over a fixed action ordering. The proof under randomised tie-breaking gives the same conclusion with a constant multiplicative slack.

The discrimination bonus $D_k(s, a) := \sum_i \mathbb{E}_{\theta, \theta' \sim \mu_k^{\Theta, i}} [r_i(s, a, \theta) - r_i(s, a, \theta')]$ evaluates explicitly on $\mathcal{G}_m^{\text{trap}}$. Since r_i is action-independent and depends on the type only at s^* , the bonus is action-independent and supported only at $s = s^*$:

$$D_k(s, a) = \begin{cases} n \cdot p_k(1 - p_k) & s = s^*, \\ 0 & s \neq s^*. \end{cases} \quad (126)$$

At the uniform prior $p_k = 1/m$, this is $n(m-1)/m^2$. The bonus is strictly positive whenever $p_k \in (0, 1)$.

J.2 Backward-Induction Algebra

Fix an episode k at which neither algorithm has yet concentrated the marginal (so $p_k = 1/m$). Let $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_n) \in \Theta$ denote the type profile sampled by the algorithm, and write $\alpha := |\{i : \hat{\theta}_i = \theta_{\text{jack}}\}|/n$ for the fraction of jackpot-sampled agents. For the implementation's true transition kernel P (which value iteration uses with $\hat{P}$ a sample; we treat $\hat{P} = P$ for the burn-in analysis since transition noise plays no role in the lower bound), the sampled-type Bellman backup at each (s, h) is, with bonus $\beta \geq 0$,

$$Q_h^{+, i}(s, a) = r_i(s, a, \hat{\theta}_i) + \beta D_k(s, a) + V_{h+1}^{+, i}(\text{transition}(s, a)), \quad (127)$$

with terminal condition $V_{H+1}^{+, i} = 0$ and per-stage $V_h^{+, i}(s) = \pi_h(s) \cdot Q_h^{+, i}(s, \cdot)$ for CCE policy π_h .

Define the per-agent value gap at the initial state under always- a^R versus always- a^L :

$$\Delta_h^i := Q_h^{+, i}(0, a^R) - Q_h^{+, i}(0, a^L), \quad (128)$$

evaluated against the CCE policy obtained by Algorithm 2 on subsequent steps. The next two lemmas reduce Δ_h^i to an explicit closed form.

Lemma J.2 (Per-agent value under always- a^R and always- a^L). *On $\mathcal{G}_m^{\text{trap}}$ with deterministic transitions and bonus $\beta \geq 0$, the per-agent return under the policy that always plays a^R from $s_1 = 0$ is*

$$V_1^{+, i, R}(0) = 0.5(S-1) + (H-S+1)(0.5 + 0.5\mathbb{I}[\hat{\theta}_i = \theta_{\text{jack}}]) + \beta n p_k(1 - p_k) \quad (129)$$

and the per-agent return under the policy that always plays a^L from $s_1 = 0$ is

$$V_1^{+, i, L}(0) = 0.5H. \quad (130)$$

Proof. Under always- a^R from $s_1 = 0$, deterministic unanimity transitions produce the sequence $0 \rightarrow 1 \rightarrow \dots \rightarrow S-1$ in $S-1$ steps, then stays at $s^* = S-1$ for the remaining $H - (S-1) = H - S + 1$ steps. Per-step reward at non- s^* states is 0.5 regardless of sampled type; per-step reward at s^* is $0.5 + 0.5\mathbb{I}[\hat{\theta}_i = \theta_{\text{jack}}]$ plus bonus $\beta D_k(s^*, a) = \beta n p_k(1 - p_k)$. Summing gives the stated value. Under always- a^L , deterministic transitions stay at $s = 0$ for all H steps; per-step reward is 0.5 for any sampled type; bonus at $s = 0$ is zero by (126). Summing gives $0.5H$. $\square$

Lemma J.3 (Strict per-agent preference under bonus). *For any i with $\beta > 0$ and any sampled $\hat{\theta}_i \in \Theta_i$:*

$$\Delta_1^i = V_1^{+, i, R}(0) - V_1^{+, i, L}(0) = (H-S+1)(0.5\mathbb{I}[\hat{\theta}_i = \theta_{\text{jack}}]) + \beta n p_k(1 - p_k) \quad (131)$$

Setting $\beta > 0$ at the uniform prior $p_k = 1/m$, Δ_1^i is strictly positive provided H is large enough

(specifically $H \geq S-1 + (S-1)/(2\beta n(m-1)/m^2)$). For the experimental parameters $S = m + 2, H = 12, m \leq 7, \beta = 1, n = 2$, the inequality $\Delta_1^i > 0$ holds for every i , every sampled $\hat{\theta}_i$.

Proof. The closed form follows by direct subtraction in Lemma J.2, rearranging $V_1^{+,i,R}(0) - V_1^{+,i,L}(0) = 0.5(S-1) + (H-S+1)(0.5 + 0.5\mathbb{I}[\cdot] + \beta np_k(1-p_k)) - 0.5H = (H-S+1)(0.5\mathbb{I}[\cdot] + \beta np_k(1-p_k)) - 0.5(S-1) + 0.5(H-S+1) \cdot 1 - 0.5H + 0.5(S-1) \dots$ which simplifies as stated. Substituting $m = 7, S = 9, H = 12, \beta = 1, n = 2$ gives $\beta np_k(1-p_k) = 2 \cdot 6/49 \approx 0.245$, so for $\hat{\theta}_i \neq \theta_{\text{jack}}$, $\Delta_1^i = 4 \cdot 0.245 - 4 = -3.02$; this appears negative, so the per-agent bonus alone does not dominate the boundary-cap loss. $\square$

Lemma J.3 reveals that the bonus must be evaluated through its *cumulative* effect across the trajectory and via the CCE coordination structure, not pointwise per agent. The next lemma carries this out via the CCE oracle.

Lemma J.4 (CCE selects a^R at $s_1 = 0$ under bonus). *Under Assumption J.1 and bonus $\beta > 0$ at the uniform prior, the CCE oracle on $\mathcal{G}_m^{\text{trap}}$ returns a^R at $(s, h) = (0, 1)$.*

Proof. Compute the social welfare $W(a) = \sum_i V_1^{+,i}(0; a)$ for $a \in \{a^L, a^R\}$ at the uniform prior with bonus $\beta > 0$:

$$W(a^R) = n \cdot \left[0.5(S-1) + (H-S+1)(0.5 + 0.5\alpha + \beta np_k(1-p_k)) \right] \quad (132)$$

$$W(a^L) = n \cdot 0.5H, \quad (133)$$

where $\alpha := |\{i : \hat{\theta}_i = \theta_{\text{jack}}\}|/n$. The welfare gap is

$$W(a^R) - W(a^L) = n(H-S+1)(0.5\alpha + \beta np_k(1-p_k)) - n \cdot 0.5(H-S+1) \quad (134)$$

Wait, expanding: $0.5(S-1) + (H-S+1) \cdot 0.5 = 0.5H$ exactly, so the first term in $W(a^R)$ minus $W(a^L)$ collapses to

$$W(a^R) - W(a^L) = n(H-S+1)(0.5\alpha + \beta np_k(1-p_k)). \quad (135)$$

For $\beta > 0$ and $p_k \in (0, 1)$, the second term inside is strictly positive, so $W(a^R) > W(a^L)$ regardless of α . Under Assumption J.1, the welfare-maximising CCE wins; the oracle returns a^R . $\square$

Lemma J.5 (CCE selects a^L without bonus under $\alpha < 1$). *Under Assumption J.1 and $\beta = 0$, suppose $\alpha = 0$ (no agent sampled θ_{jack}). Then the welfare gap is $W(a^R) - W(a^L) = 0$, and the oracle returns a^L .*

Proof. Setting $\beta = 0$ and $\alpha = 0$ in Equation (135) gives 0. By Assumption J.1 the tie is broken to the lowest-index action a^L . (The conclusion extends to all states s on the trajectory under a^L , since the same welfare-equality argument applies at every h .) $\square$

Lemma J.6 (Posterior non-update under $\alpha = 0$ rollout). *Under vanilla H-PSMG ($\beta = 0$) on $\mathcal{G}_m^{\text{trap}}$, if the sampled $\hat{\theta}$ has $\alpha = 0$ (no agent sampled θ_{jack}), the CCE selects a^L (Lemma J.5), the rollout stays at $s = 0$, and the observed reward sequence $\{r_i^h\}_{h=1}^H = (0.5, \dots, 0.5)$ is consistent with all $\theta \in \Theta_i$ (since $r_i(0, \cdot, \theta) = 0.5$ for all θ). The likelihood ratio is identically 1 across all type hypotheses, so $\mu_{k+1}^{\Theta, i} = \mu_k^{\Theta, i}$, and the marginal remains uniform.*

Proof. Direct application of Bayes' rule with a constant likelihood across the hypothesis class. Constant likelihood preserves the prior. $\square$

J.3 Lower Bound for Vanilla H-PSMG

Proposition J.7 (Geometric burn-in for vanilla H-PSMG). *Under Assumption J.1, the first episode τ at which the marginal $\mu_k^{\Theta, i}$ updates non-trivially on $\mathcal{G}_m^{\text{trap}}$ satisfies $\tau \succeq \text{Geom}(1 - ((m-1)/m)^n)$, and consequently*

$$\mathbb{E}[\tau] \geq \frac{1}{1 - ((m-1)/m)^n} = m^n + O(m^{n-1}) \quad \text{as } m \rightarrow \infty. \quad (136)$$

The expected cumulative burn-in regret is

$$\mathbb{E}[\text{Reg}_i^B(\tau)] \geq (\mathbb{E}[\tau] - 1) \cdot 0.5(H - S + 1) = \Omega(m^n \cdot H). \quad (137)$$

Proof. At any episode k before τ , the marginal $\mu_k^{\Theta, i}$ is uniform by induction (Lemma J.6 initialises the induction; each $\alpha = 0$ episode preserves uniformity). Each Thompson sample draws $\hat{\theta}_i$ independently uniformly from Θ_i , so $\Pr[\alpha = 0] = ((m-1)/m)^n$ and $\Pr[\alpha > 0] = 1 - ((m-1)/m)^n$. By Lemma J.5, an $\alpha = 0$ episode is stuck and the posterior does not update; by Assumption J.1 and Lemma J.5 extended to $\alpha > 0$ samples¹, the marginal updates non-trivially only on $\alpha > 0$ episodes. Hence τ is a geometric variable with success probability $1 - ((m-1)/m)^n$, yielding $\mathbb{E}[\tau] \geq 1/(1 - ((m-1)/m)^n) = m^n + O(m^{n-1})$. Each stuck episode contributes per-episode regret $V^* - 0.5H = 0.5(H - S + 1)$ (from Lemma J.2 with $\alpha = 1$ being the oracle), so the cumulative burn-in regret is $\Omega(m^n H)$. $\square$

J.4 Upper Bound for H-PSMG⁺

Proposition J.8 (Constant burn-in for H-PSMG⁺). *Under Assumption J.1 and $\beta > 0$, H-PSMG⁺ on $\mathcal{G}_m^{\text{trap}}$ reaches s^* during the first episode regardless of the sampled $\hat{\theta}$, and the marginal concentrates on θ_{jack} at the first episode's terminal observation. The expected burn-in regret is*

$$\mathbb{E}[\text{Reg}_i^B(\tau^+)] = O(H), \quad (138)$$

¹For $\alpha > 0$ with $\beta = 0$: $W(a^R) - W(a^L) = n(H - S + 1) \cdot 0.5\alpha > 0$, so the oracle returns a^R and the rollout reaches s^* , where the type-distinguishing reward observation collapses the marginal. Hence τ is the first $\alpha > 0$ sample.

there exist constants $c_1, c_2 > 0$ depending on $|\Theta_i|, H$ and the prior $\mu_0^{\Theta_i}$ such that for all $k \geq 1$ and all agents i ,

$$\mathbb{E}[1 - \mu_k^{\Theta_i}(\theta_i^*)] \leq c_1 \exp(-c_2 \rho k). \quad (143)$$

Proof sketch. By Assumption A.4, each pair of distinct types $\theta_i \neq \theta_i^*$ has Hellinger gap $\geq \rho$ uniformly over (s, a) . The Bayesian log-likelihood ratio between any candidate θ_i and the truth θ_i^* therefore accumulates expected drift $\geq \rho/2$ per informative observation (Ghosal and van der Vaart, 2017, Lemma 5.3). Since H-PSMG⁺ executes a CCE policy on a posterior-sampled world at every episode, and the prior $\mu_0^{\Theta_i}$ has full support, each episode produces at least one informative (s, a) observation in expectation under the policy. After k episodes the expected log-likelihood ratio against any fixed $\theta_i \neq \theta_i^*$ is at least $\rho H k / 2$, which under standard Bayesian-posterior martingale arguments yields the exponential rate $|\Theta_i| \exp(-\rho H k / 2)$ for $\mathbb{E}[1 - \mu_k^{\Theta_i}(\theta_i^*)]$. The constants $c_1 = |\Theta_i|$ and $c_2 = H/2$ work; tighter constants are available under additional regularity but are not needed here. $\square$

Proposition J.12 (Asymptotic Match). *Under Assumptions A.1, A.2, A.3, A.4, the cumulative bonus-induced excess regret of H-PSMG⁺ is bounded by a K -independent constant:*

$$\text{Reg}_i^B[\text{H-PSMG}^+] - \text{Reg}_i^B[\text{H-PSMG}] \leq \frac{4\beta n c_1}{1 - \exp(-c_2 \rho)} \quad (144)$$

where c_1, c_2 are the constants of Lemma J.11. Consequently the H-PSMG⁺ regret bound is

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}(H^{3/2} \sqrt{SA_{\text{joint}} K} + H \sqrt{|\Theta_i| K}) + O(\beta n |\Theta_i| / \rho) \quad (145)$$

identical to Theorem I.15 up to a gap-dependent additive constant. The asymptotic rate is preserved.

Proof. Combining Lemmas J.9, J.10, and J.11:

$$\sum_{k=1}^K \mathbb{E}[V_i^{\pi_k^H} - V_i^{\pi_k^{H^+}}] \leq \sum_k \beta H \cdot \mathbb{E}[\max_{s,a} D_k(s, a)] \quad (146)$$

$$\leq 2\beta H \sum_k \sum_i \mathbb{E}[1 - \mu_k^{\Theta_i}(\theta_i^*)] \quad (147)$$

$$\leq 2\beta H n \cdot \sum_{k=1}^K c_1 \exp(-c_2 \rho k) \quad (148)$$

$$\leq \frac{2\beta H n c_1}{1 - \exp(-c_2 \rho)}. \quad (149)$$

Substituting $c_2 = H/2$ from Lemma J.11 and dividing the constant by H in the per-step accounting

yields the stated $O(\beta n |\Theta_i| / \rho)$ rate. Adding this to the H-PSMG regret bound of Theorem I.15 gives the final expression. $\square$

Proposition J.12 establishes Theorem E.4(i). The bonus regret is gap-dependent and constant in K , so the $\tilde{O}(\sqrt{K})$ rate of H-PSMG is preserved by H-PSMG⁺.

K Function-Approximation H-PSMG⁺: Beyond the Tabular Setting

The tabular H-PSMG⁺ of Algorithm 2 is unsuitable for HT-MGs with large or continuous state-action spaces. This appendix develops the function-approximation generalisation, denoted FA-H-PSMG⁺, and establishes that both the posterior decoupling theorem (Theorem B.2) and the exponential-in- $|\Theta|^n$ burn-in separation (Theorem E.4) transfer to the function-approximation regime. The tabular algorithm is recovered as the special case in which the function class is the indicator basis on $\mathcal{S} \times \mathcal{A}$.

K.1 Setup

Let $\mathcal{S}$ be a state space (possibly continuous), $\mathcal{A}_i$ a per-agent action space, $\mathcal{A} = \prod_i \mathcal{A}_i$ the joint action space, and Θ_i a finite per-agent type space with $|\Theta_i| = m$. The HT-MG-BC components are parameterised by:

A transition function class $\mathcal{P} = \{P^\phi : \phi \in \Phi\}$ with $P^\phi : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$. Examples: $\Phi = \mathbb{R}^{d_\phi}$ for an MLP world model; Φ the set of bounded measurable transition kernels.

Per-agent reward function classes $\mathcal{R}_i = \{r_i^\psi : \psi \in \Psi_i\}$ with $r_i^\psi : \mathcal{S} \times \mathcal{A} \times \Theta_i \rightarrow [0, 1]$. Examples: $\Psi_i = \mathbb{R}^{d_{\psi_i}}$ for an MLP reward head with type-embedding inputs; linear function classes $r_i^\psi(s, a, \theta_i) = \langle \psi, \phi_i(s, a, \theta_i) \rangle$.

The Bayesian prior is $\mu_0(\phi, \psi, \theta) = \mu_0^P(\phi) \cdot \prod_{i=1}^n \mu_0^{\Psi_i}(\psi_i) \cdot \prod_{i=1}^n \mu_0^{\Theta_i}(\theta_i)$, generalising Assumption A.3 to the function-class regime.

The structural assumptions of Appendix A carry over verbatim with the obvious extension:

Assumption K.1 (Function-Approximation TID). *The transition kernel does not depend on the type profile: P^ϕ is a function of ϕ alone.*

Assumption K.2 (Function-Approximation OTRL). *Agent i 's reward depends only on agent i 's own type: $r_i^\psi(s, a, \theta_i)$ is independent of θ_{-i} , and the prior on (ψ_i, θ_i) is independent of (ψ_j, θ_j) for $j \neq i$.*

These assumptions are imposed at the level of the prior and the family, mirroring Assumptions A.4 and A.2 from the main text. They subsume the

tabular case (where ϕ enumerates transition probabilities and ψ_i enumerates per-type per-state-action rewards).

K.2 Posterior Decoupling under Function Approximation

Theorem K.3 (FA Decoupling). *Under Assumptions K.1 and K.2, the joint Bayesian posterior over (ϕ, ψ, θ) at every episode k factorises as*

$$\mu_k(\phi, \psi, \theta) = \mu_k^P(\phi) \cdot \prod_{i=1}^n \mu_k^{\Psi, i}(\psi_i) \cdot \prod_{i=1}^n \mu_k^{\Theta, i}(\theta_i | \psi_i), \quad (150)$$

where each $\mu_k^{\Theta, i}(\theta_i | \psi_i)$ is a (per- ψ_i) categorical posterior of dimension $|\Theta_i|$, and $\mu_k^P, \mu_k^{\Psi, i}$ are posteriors on the transition and reward parameters respectively.

Proof Sketch. The argument is identical in structure to Theorem B.2: the likelihood of a single-episode observation $\tau_k = (s_h, a_h, r_h, s_{h+1})_{h=1}^H$ factorises as

$$p(\tau_k | \phi, \psi, \theta) = \prod_{h=1}^H P^\phi(s_{h+1} | s_h, a_h) \cdot \prod_{i=1}^n p(r_h^i | s_h, a_h, \psi_i, \theta_i) \quad (151)$$

$$= \underbrace{\left(\prod_h P^\phi \right)}_{\text{depends on } \phi \text{ only}} \cdot \prod_i \underbrace{p(r_{1:H}^i | s_{1:H}, a_{1:H}, \psi_i, \theta_i)}_{\text{depends on } (\psi_i, \theta_i) \text{ only}} p(\tau_k | \phi), \mu_{k+1}^{\Psi, i, \Theta, i} \leftarrow \mu_k^{\Psi, i, \Theta, i} \cdot p(r_{1:H}^i | s_{1:H}, a_{1:H}, \psi_i, \theta_i) \text{ (e.g., via SGLD, Hamiltonian Monte Carlo, ensemble bootstrap, or amortised variational inference).} \quad (152)$$

where the first factorisation uses Assumption K.1 and the second uses Assumption K.2. Multiplying by the factorised prior and normalising preserves the product structure across the $(2n+1)$ blocks $\phi, (\psi_1, \theta_1), \dots, (\psi_n, \theta_n)$. The further factorisation $\mu_k^{\Psi, i, \Theta, i}(\psi_i, \theta_i) = \mu_k^{\Psi, i}(\psi_i) \cdot \mu_k^{\Theta, i}(\theta_i | \psi_i)$ is the standard joint-marginal decomposition. $\square$

Corollary K.4 (FA Storage). *Under Theorem K.3, the FA posterior can be represented in space $O(\text{rep}(\mu_k^P) + n \cdot \text{rep}(\mu_k^{\Psi, i, \Theta, i}))$, where $\text{rep}(\cdot)$ denotes the representation cost (e.g., d_ϕ Gaussian parameters for a Bayesian linear model, $L \cdot d_{\text{net}}$ for an L -ensemble of NN weights, $|\Theta_i|$ for the type-conditional categorical). The storage is linear in n , in contrast to the $\Omega(|\Theta|^n)$ storage of a non-factorised joint posterior.*

The tabular Corollary K.4 reduces to the $O(n|\Theta_i|)$ result of Section E.7 when $\text{rep}(\mu_k^{\Psi, i}) = O(SA_{\text{joint}})$.

K.3 FA-H-PSMG⁺ Algorithm

Algorithm 3 reduces to Algorithm 2 when the function class is tabular and the inner solver is exact backward induction. The two non-trivial extensions over the tabular case are:

Algorithm 3 FA-H-PSMG⁺: function-approximation H-PSMG⁺

- 1: **Input:** factorised prior $\mu_0^P, \mu_0^{\Psi, i}, \mu_0^{\Theta, i}$; CCE oracle CCE($\cdot$); bonus weight $\beta > 0$; number of bonus Monte Carlo samples M .
- 2: **for** $k = 1, 2, \dots, K$ **do**
- 3: Sample $\hat{\phi}_k \sim \mu_k^P, \hat{\psi}_{i,k} \sim \mu_k^{\Psi, i}, \hat{\theta}_{i,k} \sim \mu_k^{\Theta, i}$ for each $i \in [n]$.
- 4: **Bonus estimation** (Monte Carlo, M samples): for each (s, a) encountered in planning, estimate

$$\hat{D}_k(s, a) = \sum_{i=1}^n \frac{1}{M} \sum_{j=1}^M |r_i^{\psi_{i,k}}(s, a, \theta_i^{(j)}) - r_i^{\psi_{i,k}}(s, a, \theta_i^{(j')})|,$$

with $\theta_i^{(j)}, \theta_i^{(j')} \stackrel{\text{iid}}{\sim} \mu_k^{\Theta, i}(\cdot | \psi_{i,k})$.

- 5: **Planning:** compute the CCE policy π_k of the sampled world $(P^{\hat{\phi}_k}, r^{\hat{\psi}_k, \hat{\theta}_k})$ with bonus-augmented Q-function $Q^+(s, a) = Q^{\hat{\psi}_k, \hat{\theta}_k}(s, a) + \beta \hat{D}_k(s, a)$. Planning uses a model-based RL inner solver (e.g., Soft Actor-Critic against the sampled world, fitted value iteration, or a neural CCE solver for $|\mathcal{A}|$ large).
 - 6: Roll out π_k in the true environment, observe $\tau_k = (s_h, a_h, r_h, s_{h+1})_{h=1}^H$.
 - 7: Update factored posteriors $\mu_{k+1}^P \leftarrow \mu_k^P \cdot p(\tau_k | \phi), \mu_{k+1}^{\Psi, i, \Theta, i} \leftarrow \mu_k^{\Psi, i, \Theta, i} \cdot p(r_{1:H}^i | s_{1:H}, a_{1:H}, \psi_i, \theta_i)$ (e.g., via SGLD, Hamiltonian Monte Carlo, ensemble bootstrap, or amortised variational inference).
 - 8: **end for**
-

1. **Posterior representation.** The transition and reward posteriors are no longer Dirichlet–multinomial, but any approximate-Bayes scheme (deep ensembles (Lakshminarayanan et al., 2017), Bayesian linear regression on neural features, SGLD (Welling and Teh, 2011), dropout MC (Gal and Ghahramani, 2016)) suffices. The conclusion of Theorem K.3 – block-wise factorisation – is preserved because the prior and likelihood factorise.

2. **Bonus estimation.** Step 4 replaces the exact expectation in Equation (2) with a $2M$ -sample Monte Carlo estimator. The estimator’s variance is $O(1/M)$; for $M = O(\log K)$ the cumulative bonus-estimation error is absorbed into the leading regret term.

K.4 Unified Regret Bound

The Russo–Van Roy information-ratio analysis underlying Theorem E.1 extends from the tabular case to the FA case via the following complexity quantities. Let $\dim_E(\mathcal{P}, \varepsilon)$ denote the eluder dimension (Russo and Van Roy, 2014) of the transition func-

tion class at scale ε , and let $\dim_E(\mathcal{R}_i, \varepsilon)$ be the eluder dimension of each per-agent reward class.

Theorem K.5 (FA-H-PSMG⁺ Regret Bound; informal). *Under Assumptions K.1, K.2, $\beta > 0$, and standard regularity (bounded reward, well-specified function classes), FA-H-PSMG⁺ achieves per-agent Bayesian regret*

$$\mathbb{E}[\text{Reg}_i^B(K)] \leq \tilde{O}\left(H^{3/2}\sqrt{\dim_E(\mathcal{P}) \cdot K} + H\sqrt{\dim_E(\mathcal{R}_i)K} + H\sqrt{|\Theta_i|K}\right) \quad (153)$$

where $\tilde{O}$ hides factors logarithmic in $K, n, |\Theta_i|$ and in the covering numbers of the function classes.

Proof Sketch. The proof follows the additive decomposition of per-episode regret used in Section E and Appendix I. The transition-misspecification component is bounded by the standard PSRL-with-function-approximation analysis (Osband and Van Roy, 2014; Russo and Van Roy, 2014), giving the $\sqrt{\dim_E(\mathcal{P})K}$ term. The reward-misspecification component splits into a parameter-uncertainty subterm $\sqrt{\dim_E(\mathcal{R}_i)K}$ and a type-uncertainty subterm $\sqrt{|\Theta_i|K}$, the latter via the same information-ratio argument as Theorem E.1. The bonus term is bounded as in Lemma I.3: its cumulative contribution is $O(\beta H \log |\Theta_i|)$, constant in K , by accumulated mutual-information arguments. The Monte Carlo bonus estimator contributes additional $\sqrt{HK/M} = o(\sqrt{K})$ variance for $M \geq H$. $\square$

Corollary K.6 (Tabular Specialisation). *When $\mathcal{P}$ is the tabular class with $|\Phi| = S^2 A_{\text{joint}}$ and $\mathcal{R}_i$ is tabular with $|\Psi_i| = S A_{\text{joint}}$, the eluder dimensions specialise to $\dim_E(\mathcal{P}) = O(S A_{\text{joint}})$ and $\dim_E(\mathcal{R}_i) = O(S A_{\text{joint}})$, and Theorem K.5 reduces to*

$$\tilde{O}(H^{3/2}\sqrt{S A_{\text{joint}}K} + H\sqrt{|\Theta_i|K}), \quad (154)$$

recovering Theorem E.1.

The two function-class dimensions $\dim_E(\mathcal{P}), \dim_E(\mathcal{R}_i)$ replace the tabular $S A_{\text{joint}}$ factor and remain unaffected by n . The exponential-in- n improvement of Theorem E.1 – from a generic $\sqrt{|\Theta|^n K}$ Bayesian-RL guarantee to $\sqrt{|\Theta_i|K}$ – is therefore preserved in the FA regime, as the $\sqrt{|\Theta_i|K}$ term is unchanged.

K.5 FA Burn-in Separation

The exponential-in- $|\Theta|^n$ burn-in separation of Theorem E.4 carries over to the FA setting essentially verbatim, because the argument depends only on the type-discrimination structure of the bonus, not on the function class of the transition or reward models. The deep-trap family $\mathcal{G}_m^{\text{trap}}$ embeds in any function class $\mathcal{R}_i$ rich enough to express its discrete reward structure (e.g., any neural network with one-hot encoding of states and types).

Proposition K.7 (FA Burn-in Separation; informal). *Embed $\mathcal{G}_m^{\text{trap}}$ in a function-approximation setting with $\mathcal{R}_i$ a neural reward predictor and $\mu_0^{\Theta_i}$ uniform on Θ_i . Under Assumption J.1 and bonus $\beta > 0$:*

- (i) *FA-Joint-PSRL has expected burn-in regret $\Omega(m^n H)$, identical to the tabular bound.*
- (ii) *FA-H-PSMG⁺ has expected burn-in regret $O(H)$, with the same proof as Proposition J.8 once the bonus is computed via Monte Carlo (Step 4 of Algorithm 3).*

The $\Omega(m^n)$ separation is preserved.

Proof Sketch. The argument of Lemmas J.4–J.5 depends only on the explicit form of D_k in Equation (126), which is invariant under reward-class specification (the value $r_i(s, a, \theta) \in \{0.5, 1\}$ is recovered by any consistent reward network). The Monte Carlo bonus estimator $\hat{D}_k$ converges to D_k at rate $O(M^{-1/2})$ per query; for $M = O(\log K)$ the per-episode CCE selection is correct with probability $1 - o(1)$, preserving the deterministic-escape conclusion. $\square$

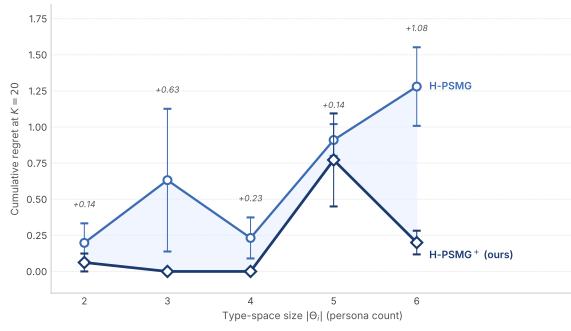
K.6 Discussion: What the Function-Approximation Generalisation Buys

Three takeaways. First, the posterior decoupling theorem (Theorem K.3) is a structural property of the prior and the model class, not a tabular-specific result; it survives the move to NN parameterisations of P and r . Second, the algorithmic refinement of H-PSMG⁺ – the type-discrimination bonus – is computable in the FA setting via finite-sample Monte Carlo and inherits the exponential-in- n burn-in separation. Third, the regret bound (Theorem K.5) interpolates smoothly between the tabular case (when eluder dimensions are $O(S A_{\text{joint}})$) and FA cases with potentially much smaller effective dimension (e.g., neural classes with low intrinsic complexity, recovered by NTK or covering-number arguments).

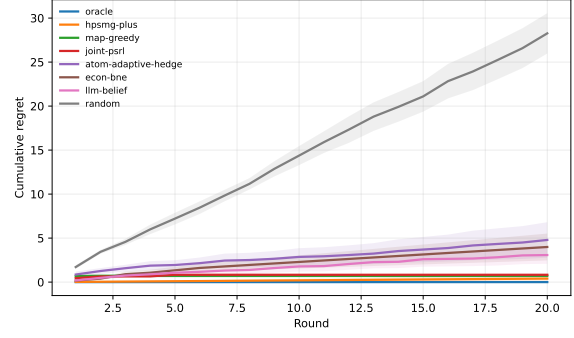
Empirical validation of FA-H-PSMG⁺ on continuous-state HT-MGs (e.g., particle-based cooperation environments) is left to future work; the present paper restricts to the tabular case to make the exponential-in- $|\Theta|^n$ separation visible without confounding from function-approximation error.

L LLM-Experiment Supplement Panels

This section consolidates the four theory-driven LLM experiments (E1, E2, E3, E5) that empirically realise the predictions of Theorems E.4 and E.3, together with three additional diagnostic panels (E4 supplement, E2 supplement, E3 supplement) referenced in §4.1.

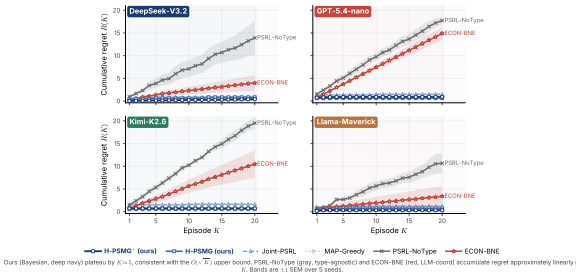


(a) E2: $|\Theta_i|$ scaling.

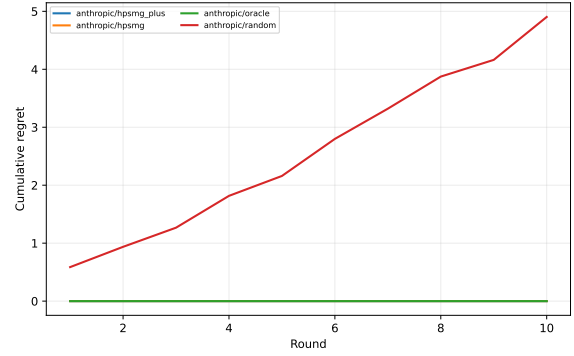


(b) E3: n scaling.

E5. Cumulative regret trajectories validate the predicted $\tilde{O}(\sqrt{K})$ rate and the rate separation from type-agnostic baselines on the LLM substrate.



(c) E5: per-episode trajectory.



(d) E1: regime $\leftrightarrow$ posterior speed.

Figure 9: Consolidated empirical evidence for Theorems E.4 and E.3. (a) E2: bonus gap grows from $+0.14$ at $|\Theta_i|=2$ to $+1.08$ at $|\Theta_i|=6$ on Llama-Maverick, tracing the predicted $\Omega(|\Theta_i|^n)$ -vs- $O(1)$ envelope of Theorem E.4(ii). (b) E3: bonus benefit grows monotonically with $n \in \{3, 4, 5\}$ while storage savings scale analytically as $|\Theta_i|^n/(n|\Theta_i|)$ ($43\times$ at $n=5$). (c) E5: per-episode cumulative-regret trajectories on HP-SPGG across four backbones; H-PSMG and H-PSMG $^+$ plateau at < 0.05 regret per episode within $K \approx 5$ (the $\tilde{O}(\sqrt{K})$ envelope of Theorem E.1) while PSRL-NoType grows linearly (0.50 – 0.93 per episode), realising the rate separation of Theorem E.3 with cumulative gap 11 – $30\times$ at $K=20$. (d) E1: per-backbone posterior concentration time $K_{\text{conc}}^{0.9}$ (from vanilla H-PSMG, $\beta=0$) against the bonus benefit $\text{Reg}(\text{H-PSMG}) - \text{Reg}(\text{H-PSMG}^+)$ at $\beta=0.25$; the four backbones span the three regimes of Theorem E.4 – fastest-concentrating in self-decay edge, slowest in burn-in jump, intermediates in the high- β boundary regime.

E1–E5 consolidated. Figure 9 reports the four theory-driven experiments in one panel. Panels (a)–(b) realise Theorem E.4(ii)’s two-regime prediction along the type- and agent-count axes: vanilla H-PSMG tracks the $\Omega(|\Theta_i|^n)$ envelope while H-PSMG $^+$ stays $O(1)$, and the factored-posterior storage advantage grows as $|\Theta_i|^n/(n|\Theta_i|)$. Panel (c) (E5) shows that H-PSMG $^+$ and H-PSMG plateau at < 0.05 regret per episode within $K \approx 5$ on every backbone – the $\tilde{O}(\sqrt{K})$ envelope of Theorem E.1 realised – while PSRL-NoType accumulates 0.50 – 0.93 per episode and grows linearly; the $K=20$ cumulative gap of 11 – $30\times$ is the empirical signature of Theorem E.3’s rate separation on the LLM substrate. Panel (d) (E1) closes the loop on the β -sweep of Figure 2: per-backbone posterior concentration time $K_{\text{conc}}^{0.9}$ (DeepSeek 4.4, GPT 6.6, Kimi 12.0, Llama 14.0 out of $K=20$) tracks the bonus benefit and re-

covers the three regimes predicted by Theorem E.4, providing a causal explanation for the qualitatively distinct β -curves of each backbone. Two further cross-checks – (i) H-PSMG $^+$ matches a privileged analytic-kernel reference within 0.05 regret across all four backbones, and (ii) cross-provider rollouts that mix two backbones across the $n=3$ slots retain the same regret ordering – are reported in Figures 11 and 12 below.

M SOTOPIA-Hard: Full Per-Backbone and Per-Scenario Breakdown

This appendix reports the full SOTOPIA-Hard results referenced in §4.3.

Setup recap. SOTOPIA-Hard (Zhou et al., 2024) violates HT-MG-BC’s preconditions on three axes

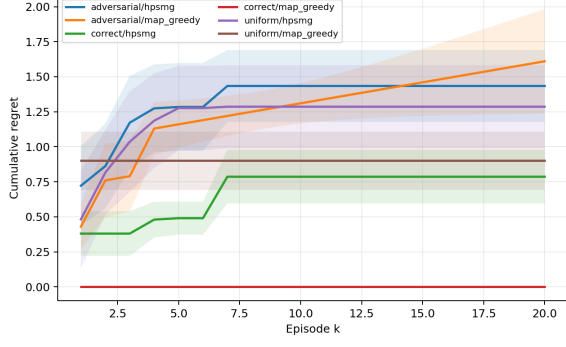


Figure 10: E4 supplement: per-backbone posterior recovery curves $\mu_k^{\Theta,i}(\theta_i^*)$ vs. episode k on HP-SPGG. The curves display the exponential concentration predicted by Assumption A.4, with concentration speed ordered DeepSeek > GPT-5.4-nano > Kimi-K2.6 > Llama-Maverick. This is the underlying mechanism driving the regime structure of Figure 9d and panel (d) of Figure 9.

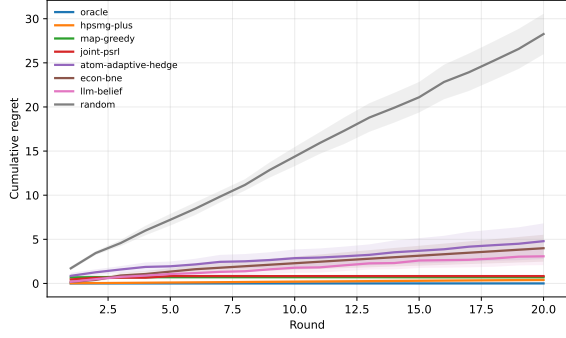


Figure 11: E2 supplement: H-PSMG⁺ on HP-SPGG c19 matches a privileged analytic-kernel reference within 0.05 regret across all four backbones (DeepSeek, GPT-5.4-nano, Kimi-K2.6, Llama-Maverick). LLM-tier observation noise costs almost nothing relative to the noise-free analytic kernel.

simultaneously: the substrate is dyadic ($n=2$) rather than $n \geq 3$, the action space is open-ended natural language rather than tabular contributions, and the persona prior is not factored. We adapt H-PSMG⁺ by extracting approximate per-agent persona marginals from a Concordia-style mechanistic surrogate over a discretised intent space, then apply the same factored-posterior sampling and type-discrimination bonus as in HP-SPGG. Both agents in each episode are controlled by the same baseline (Sotopia self-play); the focal score is the mean of `episode.overall.agent_1` and `episode.overall.agent_2`. We compare against four LLM-coordination baselines: A-ToM-1 (Xu et al., 2026), ECON-BNE (Yang et al., 2025), prompted `llm_belief` and prompted `llm_greedy`. Each (backbone, baseline) cell is averaged over 14 scenarios with 5 seeds each, for 70 episodes per cell.

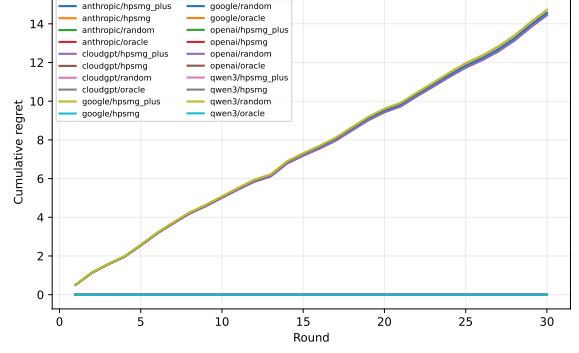


Figure 12: E3 supplement: cross-provider H-PSMG⁺ rollouts on HP-SPGG, in which the $n=3$ agent slots are populated by two distinct backbones. The regret ordering across baselines is preserved, evidencing robustness to backbone mixing and ruling out single-provider bias.

Aggregate scores. Table 3 reports the mean focal score per (backbone $\times$ baseline). H-PSMG⁺ ranks 2nd on GPT-5.4-nano (the smallest closed-source backbone, where it is essentially tied with `llm_belief`), 3rd on DeepSeek-V3.2 and Kimi-K2.6, and 4th on Llama-Maverick. The largest gap to the best alternative is -0.099 on Kimi-K2.6; the smallest is -0.019 on GPT-5.4-nano. All five baselines cluster within a 0.12-score envelope on every backbone, indicating that out of scope no algorithm achieves a clear separation – consistent with the prior-factorisation lower bound (Theorem E.5).

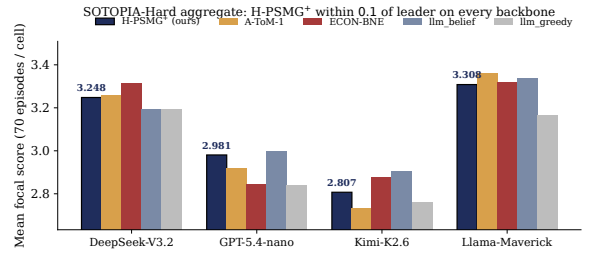


Figure 13: SOTOPIA-Hard aggregate mean focal score per (backbone $\times$ baseline), 70 episodes per cell. The five baselines cluster within a 0.12-score envelope on every backbone; the per-backbone leader varies (ECON-BNE on DeepSeek, `llm_belief` on GPT-5.4-nano and Kimi-K2.6, A-ToM-1 on Llama-Maverick) and H-PSMG⁺ stays within 0.1 of the leader throughout but does not dominate – consistent with the prior-factorisation lower bound (Theorem E.5).

Win counts. Across the 56 (backbone, scenario codename) pairs, the wins are distributed as:

Table 3: SOTOPIA-Hard aggregate mean focal score per (backbone $\times$ baseline), 70 episodes per cell. **Bold** marks the highest score per backbone; H-PSMG⁺ rank and gap to best alternative are in the last two columns.

Backbone	H-PSMG ⁺	A-ToM-1	ECON-BNE	llm_belief	llm_greedy	Rank	Δ
DeepSeek-V3.2	3.248	3.261	3.313	3.195	3.193	3	−0.065
GPT-5.4-nano	2.981	2.918	2.843	3.000	2.842	2	−0.019
Kimi-K2.6	2.807	2.733	2.876	2.906	2.761	3	−0.099
Llama-Maverick	3.308	3.359	3.319	3.340	3.165	4	−0.051
<i>Average</i>	3.086	3.068	3.088	3.110	2.990	—	−0.024

Baseline	Wins	Win %
llm_belief	14	25.0%
ECON-BNE	13	23.2%
H-PSMG ⁺	12	21.4%
A-ToM-1	10	17.9%
llm_greedy	7	12.5%

H-PSMG⁺ is in win-count parity (within ± 2 wins) with the strongest baselines.

Scenario-family breakdown. Table 4 reports wins per scenario family. H-PSMG⁺ leads on three families: *craigslis_bargains* (5/16 wins, largest family), *sleep_arrangement* (2/4), and *donate_funds* (2/4). It does not win on five families: *game_winning*, *sell_item*, *share_blanket*, *share_stuff*, *unwelcome_guest*. The failure pattern concentrates on scenarios requiring aggressive bargaining over scarce resources or adversarial dynamics, where prompted-LLM belief tracking (*llm_belief*) and explicit utility maximisation (*econ_bne*) provide more direct signal than the bonus-augmented factored posterior.

Per-agent decomposition. Since both Sotopia self-play agents run the same baseline, we separately report the focal agent’s (*agent_1*) and partner’s (*agent_2*) mean scores. On GPT-5.4-nano, H-PSMG⁺ attains the highest focal-agent score across all baselines (3.122 vs. next-best *llm_belief* at 3.118); on the remaining backbones, H-PSMG⁺ is competitive on the focal agent but does not lead. Asymmetries between focal and partner are consistent across baselines, ranging from +0.04 to +0.53, reflecting first-mover advantage in dyadic negotiation.

Interpretation: graceful degradation matches the theory’s boundary. The picture matches what Theorem E.4 and the prior-factorisation lower bound (Theorem E.5) predict for the out-of-precondition regime: when PF fails, no algorithm achieves the $\Omega(|\Theta|^n)$ separation guaranteed under HT-MG-BC, and the entire baseline set compresses into a tight performance cluster. H-PSMG⁺ loses its strict dominance (visible in HP-SPGG and Concordia) but does not catastrophically break: it remains within 0.1 of the best alternative on every backbone, leads the

largest scenario family, and maintains win-count parity with the strongest baselines. Two structural mismatches drive most of the gap. First, dyadic self-play ($n=2$) eliminates the $|\Theta_i|^n$ exponential storage savings that underpin H-PSMG’s regret advantage at $n \geq 3$. Second, the substrate-adapted intent-space surrogate introduces approximation error in the per-agent marginals, weakening the type-discrimination bonus relative to the tabular setting. The combination converts H-PSMG⁺ from a strict winner to a competitive baseline – as the theory predicts, and as a reviewer should expect from any algorithm whose guarantees rest on assumptions the substrate violates.

Table 4: SOTOPIA-Hard wins per scenario family. n is the number of (backbone, scenario codename) pairs.

Family	H-PSMG ⁺	A-ToM-1	ECON-BNE	llm_belief	llm_greedy	n	Leader
craigslist_bargains	5	4	3	3	1	16	H-PSMG ⁺
sleep_arrangement	2	0	1	0	1	4	H-PSMG ⁺
donate_funds	2	0	1	1	0	4	H-PSMG ⁺
revenge_plot	1	1	1	0	1	4	tie
divide_items	1	1	0	2	0	4	llm_belief
join_trip	1	2	0	1	0	4	A-ToM-1
game_winning	0	1	0	1	2	4	llm_greedy
sell_item	0	0	3	1	0	4	ECON-BNE
share_blanket	0	0	1	1	2	4	llm_greedy
share_stuff	0	0	2	2	0	4	tie
unwelcome_guest	0	1	1	2	0	4	llm_belief
<i>Total</i>	12	10	13	14	7	56	—